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Bachelor Thesis

Topic:

**Effects of prior distribution selection on stable diffusion
models for single-cell data**

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Data Science

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Abstract

The increasing availability of single-cell RNA sequencing (scRNA-seq) data has spurred interest in generative models that can simulate high-fidelity synthetic cells to aid in data augmentation, benchmark studies, and biological discovery. Diffusion-based generative models have recently emerged as a powerful alternative to traditional architectures such as Variational Autoencoders (VAEs) and Generative Adversarial Networks (GANs), particularly due to their stability and sample quality. However, the role of prior distribution choice in the performance of these models remains underexplored, especially in the context of biological data.

This thesis investigates how different prior distributions, specifically Gaussian, Log-normal and Student's t , impact the training dynamics and generative capabilities of stable diffusion models tailored for single-cell data. Using benchmark datasets such as Tabula Muris, we analyze the effect of prior selection on both model convergence and the statistical fidelity of generated cells. Evaluation is conducted using statistical metrics and nonstatistical metrics, which quantify the similarity between real and synthetic data distributions.

Our findings demonstrate that the choice of prior can significantly influence model behavior and the quality of generated data. This study contributes to a deeper understanding of how probabilistic assumptions in generative modeling affect downstream biological interpretations and offers practical guidance for designing future models in single-cell data simulation.

Keywords: Generative AI, Stable Diffusion model, prior distribution, scRNA-seq data analysis

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Acronyms

AI Artificial Intelligence.

DL Deep Learning.

DNA Deoxyribonucleic acid.

ED Energy Distance.

GANs Generative Adversarial Networks.

ML Machine Learning.

MMD Maximum Mean Discrepancy.

PCA Principal Component Analysis.

RNA Ribonucleic acid.

scRNA-seq single cell RNA sequencing.

UMAP Uniform Manifold Approximation and Projection.

VAE Variational Autoencoders.

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Chapter 1

Introduction

Cells are the fundamental units of all living organisms, providing structure, converting nutrients to energy, and performing specialized functions. Though varied in shape and size (e.g., skin cells and nerve cells), all cells contain the same hereditary material, or Deoxyribonucleic acid (DNA). This DNA is organized into genes, which serve as templates for protein production—a process known as the central dogma of molecular biology. Proteins form body structures and regulate cellular processes.

The central dogma involves transcription, where DNA is copied into Ribonucleic acid (RNA), and translation, where RNA is converted into amino acid sequences. This overall process, from gene to functional product, is called gene expression. Only a subset of genes is active at any given time, and these varying gene expression profiles explain the diversity of cell types and their development within an organism.

1.1 Context and Motivation

Understanding gene activity in individual cells is key to uncovering cellular diversity and function. Advances in single-cell RNA sequencing (scRNA-seq) now allow detailed profiling of gene expression at cellular resolution, offering meaningful insight into cellular heterogeneity, development, and disease [41, 20]. However, this type of data is sparse, noisy and high-dimensional, posing significant challenges for modeling. Moreover, it is costly to produce a single cell dataset, which satisfy the research requirements.

To address these challenges, deep generative models such as Variational Autoencoders (VAEs) and Generative Adversarial Networks (GANs) have been leveraged to simulate realistic single-cell profiles and capture complex data distributions [22, 23]. More recently, diffusion-based generative models have demonstrated superior performance in other domains

such as image synthesis, due to their stable training and ability to generate high-quality samples [13, 15]

A critical but often overlooked component in generative modeling is the choice of prior distribution, which can substantially affect model learning and data consistency. While most models adopt Gaussian prior by default, this assumption may not align with the skewed and overdispersed nature of gene expression data, which often follows a lognormal-like distribution [10]. Thus, understanding how prior distribution selection influences the performance of diffusion-based models is crucial for advancing generative modeling in single-cell biology.

1.2 Objectives

The goals of this thesis is to investigate the effects of prior distribution selection, which are Gaussian, Log-normal, on the training efficiency and generative performance of stable diffusion models applied to scRNA-seq data. Then quantifying the similarity between real and generated data using robust statistical distance metrics such as Energy Distance [40], Maximum Mean Discrepancy (MMD) [9] and Spearman's Rank Correlation Coefficient (Spearman SSC) together with non-statistical metrics for instance Uniform Manifold Approximation and Projection (UMAP) plot and KNN-AUC score. To compare different prior settings, this thesis also adopt baseline generative models such as VAEs [17] and GANs [8].

We will construct the thesis as below:

1.3 Thesis Structure

- Chapter 1 **Introduction**: this chapter gives context and motivation and states the main objectives for this project.
- Chapter 2 **Related Work**: this chapter summarizes all the relevant research that serves as a foundation for our project.
- Chapter 3 **Background**: this chapter provides fundamental explanations about the concepts and model using in this project, starting from Biological background, Mathematical background and ML/DL background.
- Chapter 4 **Material & Methodology**: this chapter describes the data resource, pipeline using to implement this project and the evaluation methods.

- Chapter 5 **Experiment**: this chapter reports the project training setup and results of the experiment.
- Chapter 6 **Conclusion & Discussion**: this chapter summarize the project conclusion and provide further discussion on this project.

Chapter 2

Related Work

2.1 Diffusion Models: Fundamentals and Recent Advances

Diffusion models have emerged as powerful generative models, initially introduced by Sohl-Dickstein et al. (2015) [37]. These models learn to reverse a gradual noising process, starting from random noise to generate realistic data points. Significant advancements were made by Ho et al. (2020) [13], who introduced denoising diffusion probabilistic models (DDPMs), substantially improving generation quality and training efficiency. Further developments demonstrated that diffusion models could outperform Generative Adversarial Networks (GANs) in image synthesis tasks (Dhariwal & Nichol, 2021; Song et al., 2021) [30, 38]. More recently, researchers have adapted these models for complex biological data, specifically single-cell RNA sequencing (scRNA-seq), showcasing their versatility and effectiveness in diverse domains (Luo et al., 2024; Xu et al., 2022) [25, 49]

The selection of prior distributions critically influences generative models. Priors shape latent spaces in variational frameworks such as variational autoencoders (VAEs), impacting the diversity and fidelity of generated data (Kingma & Welling, 2014) [17]. These models traditionally employ standard Gaussian priors due to computational convenience. However, recent studies have explored alternative distributions, including truncated normal, log-normal, and gamma distributions, demonstrating that different priors significantly affect convergence, stability, and generation quality (Yang et al., 2023; Nichol & Dhariwal, 2021)[50, 30]. Song and Ermon (2021) [38] provided empirical insights suggesting that the choice of prior could markedly influence training dynamics and the quality of generated samples.

2.2 Single-cell RNA-seq Data Generation

Generating synthetic scRNA-seq data poses unique challenges due to inherent characteristics like high dimensionality, sparsity, and zero-inflation (Luecken & Theis, 2019) [24]. Traditional deep generative frameworks, including scVI (Lopez et al., 2018) [22] and scGen (Lotfollahi et al., 2019) [23], have shown promise but face limitations in capturing the full complexity of scRNA-seq distributions. Recent adaptations of diffusion-based methods have shown potential to better address these challenges, leveraging their strong capacity to model intricate distributions and maintain data fidelity (Xu et al., 2022; Luo et al., 2024) [48, 25].

Evaluating synthetic single-cell data quality involves quantitative and qualitative metrics. Statistical metrics such as Energy distance, Maximum Mean Discrepancy (MMD), Spearman SSC, and KNN-AUC provide robust numeric evaluations of how closely synthetic data matches real data distributions (Lopez et al., 2018; Xu et al., 2022) [22, 48]. Additionally, visualization methods such as Uniform Manifold Approximation and Projection (UMAP) offer qualitative insights into the preservation of local and global structures, enabling intuitive assessments of data similarity and biological relevance (McInnes et al., 2018) [28].

2.3 Research Gap and Contribution of the Thesis

Despite significant progress, the specific impact of prior distribution choice on diffusion-based synthetic data generation for scRNA-seq remains underexplored. Most studies default to standard Gaussian priors without systematically comparing alternative distributions. This thesis explicitly addresses this gap by performing a comprehensive comparative analysis of Gaussian, log-normal, and Student's t priors. It evaluates how each prior affects model performance, stability, and biological relevance, providing critical insights and recommendations for prior selection in future generative modeling efforts for single-cell transcriptomics.

Chapter 3

Background

This section aims to provide a comprehensive overview about the background in which this thesis is based on. Starting from a biological background with an overview of the single-cell data and its application, following with a mathematical background in probability distributions and statistic measurement which is the foundation for the third part about generative model and Machine Learning/ Deep Learning method integrated in this thesis scope.

3.1 Biological background

3.1.1 Overview of single cell data

- Single cell data originally coming from High-throughput Sequencing methods - Bulk RNA sequencing and Single-cell RNA sequencing, which are revolutionary technologies that allows us to analyze gene expression at the resolution of multiple cells or a single cell. Traditional bulk RNA sequencing measures average gene expression across a large population of cells. While this is informative at a general level, it masks the variability and distinct functional states that exist between individual cells. In contrast, single-cell RNA sequencing (scRNA-seq) provides a high-resolution view of the transcriptome at the level of individual cells, enabling the detection of cell-to-cell heterogeneity. By capturing heterogeneity across cells, researchers can uncover developmental pathways, disease progression mechanisms, and cell-specific responses to external stimuli, which are often masked in bulk data [43].
- Several platforms have been developed for scRNA-seq, each with unique strengths in throughput, sensitivity, and coverage. Popular technologies include 10x Genomics

Chromium, which uses microfluidic droplet-based approaches; SMART-seq, known for full-length transcript capture; and Drop-seq, which allows scalable profiling at a relatively low cost [52, 26]. These platforms typically generate sparse count matrices of gene expression per cell, often requiring sophisticated computational approaches for downstream analysis.

The structure of a single-cell data is illustrated in the below diagram with 4 main components:

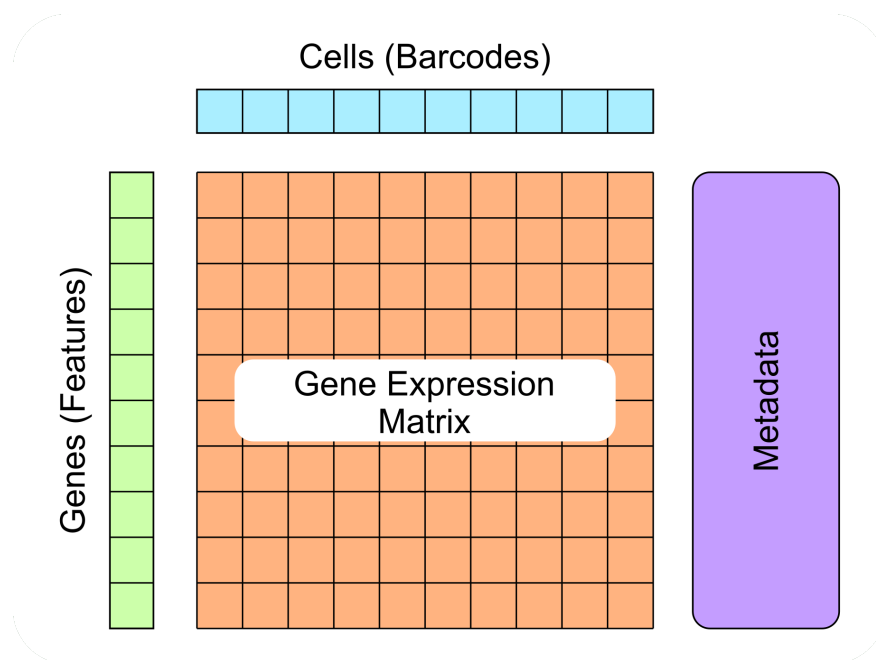


Fig. 3.1 Gene expression diagram [1]

- **Gene expression matrix:** this is the core component of single-cell data, the matrix represented gene expression values for each gene in the sequenced cells. Usually, each row can be the genes and each columns can be the cells, although this is not always the case and depends on the user settings, some matrices need to be transposed before processing furthermore. This matrix can be in the form of full (one large table where every entry contains a value) or sparse (remove all values that contain no gene expression). A sparse matrix is more preferable to stored information efficiently.
- **Cell metadata (barcodes):** this including the cell identifiers/barcodes, data on the origin of each cell, and quality control metrics received from analysis tools.
- **Gene metadata (features):** this data usually referred as feature when stored in a file, containing the gene identifiers/IDs, expressions metrics and quality control metric.

- Unstructured metadata: this data field is not associated with any individual cell or gene but belongs to the whole data, for instance, time information (when the cells were sequenced), platform of sequence, source of the cell samples, etc.

Single-cell data can have several formats and for each format, a programming language and packages are required. Many public data sources available, which provided reusable data such as: NCBI, CellXGene Collection and Human Cell Atlas: Data Explorer. Some commonly used data format are:

- Tabular: the most basic form where we stored all data in a single file (gene identifiers, cell identifiers, expression values). The file extensions usually: *.csv*, *.tsv*, *.txt*.

GENE	ACAGATCACTCA		TAACTGATCATA		CGTCCAGTTGTT		CATTTATGTGCC	
A1BG	0	0	0	0	0	0	0	0
A1BG-AS1		0	0	0	0	0	0	0
A1CF	0	0	0	0	0	1	0	0
A2M	0	0	0	0	0	1	0	0
A2M-AS1	0	0	0	0	0	0	0	0
A2ML1	0	0	0	0	0	0	0	0
A4GALT	0	1	0	0	0	0	1	0
AAAS	2	0	0	1	1	0	1	1
AACS	0	4	0	2	0	1	1	0
AADAT	0	0	0	1	0	0	0	0
AAED1	1	1	2	0	0	0	1	1
AAGAB	1	0	1	1	0	0	0	2
AAK1	0	1	0	0	0	0	0	0
AAMDC	0	2	2	1	2	0	2	0
AAMP	1	7	1	2	1	0	0	3
AAR2	0	0	1	0	1	0	0	1
AARD	0	0	0	0	0	0	0	0
AARS	3	12	2	5	5	0	1	1
AARS2	1	0	0	0	0	0	0	0

Fig. 3.2 Tabular data example [1]

- AnnData: this is a Python-based single-cell object built on the HDF5 format (use with the primary library Scanpy) and specially designed for matrix-like data. The file extension is *.h5ad*.

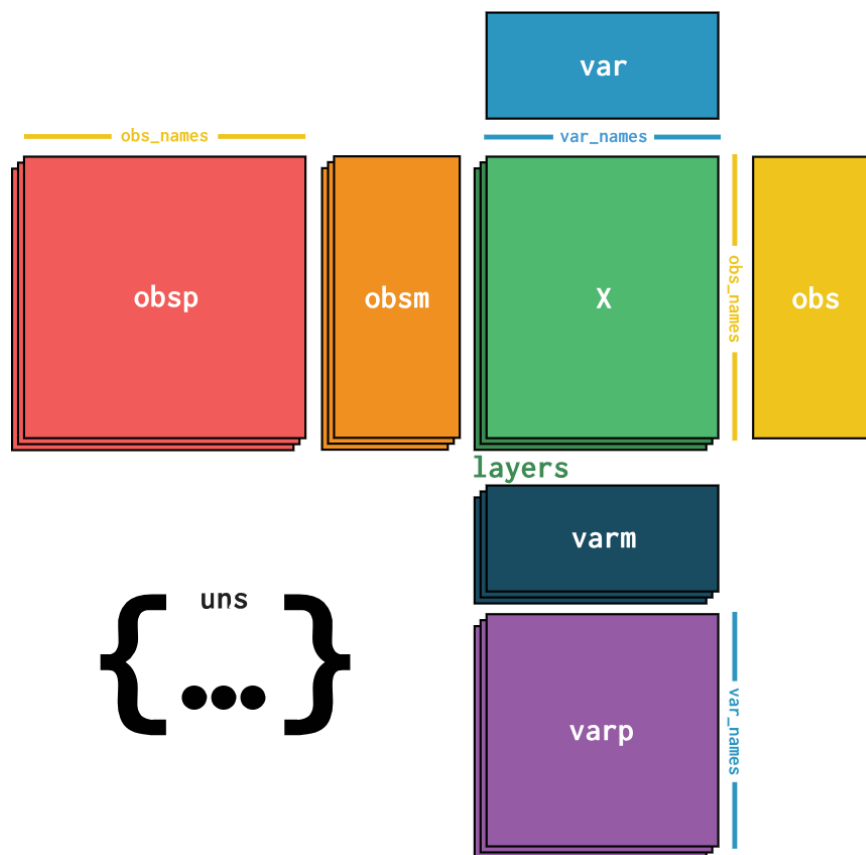


Fig. 3.3 AnnData object example [1]

- Seurat: this is an R-based format that is commonly used with Seurat package - a software package that contains various tools for single-cell data. The file extensions usually: `.rds`, `.rda`, `.rdata`.

Although tabular format are easy to read, the limitations appear when you want to stored additional metadata outside of the gene or cell identifiers. Therefore, AnnData or Seurat can be the ideal candidate for handling sparsity, unstructured data and many level of metadata coming from cells or genes.

For simplifying the data transfer process and memory storage, single-cell data usually stored in a compressed format, for example, `.zip`, `.gz` and should be unzipped before processing. When the data is stored in multiple files/folders they maybe organized into a single file called *tarball* which can be extracted with TAR tool. It is also common for multiple files/folders to be both compressed and organized into a single file, this usually have the extension `.tar.gz`. In order to used, we need to both unzipped and extracted beforehand.

The insights coming from scRNA-seq technologies can be used for multiple purposes, from cell type exploration to drug discovery, which are described in the section below.

3.1.2 Applications of single-cell data

- Cell type discovery: scRNA-seq has facilitated the identification of novel and rare cell populations across tissues and organisms. For instance, large-scale atlases like the Tabular Muris have systematically profiled thousands of cells across mouse organs to build reference maps of cellular identity [35].
- Developmental biology: Time-course and trajectory analysis using scRNA-seq have unveiled lineage differentiation paths in developmental systems, such as hematopoiesis or neural development. Pseudotime algorithms and RNA velocity further allow the inference of dynamic transcriptional changes [19].
- Disease characterization: scRNA-seq is revolutionizing cancer research by mapping intratumoral heterogeneity, identifying malignant subclones, and revealing therapy-resistant populations [32]. In immunology, it enables profiling of immune repertoires and their activation states during infection or autoimmunity [46].
- Perturbation studies: Single-cell data has been used to examine cellular responses to genetic or pharmacological perturbations. This is particularly useful in drug discovery and functional genomics, where understanding heterogeneous responses is critical [6].

However, due to special characteristics, the modeling process for single cell data can meet some challenges.

3.1.3 Challenges in Modeling scRNA-seq data

- High dimensionality: Each cell is described by thousands of genes, resulting in high-dimensional and sparse data that can be difficult to interpret and model effectively.
- Sparsity and dropout: Due to technical limitations, many transcripts are missed during sequencing, creating excess zero values. These dropout events pose significant challenges to statistical inference and generative modeling [16]. Traditional models based on Gaussian assumptions fail to capture the zero-inflated, overdispersed nature of scRNA-seq data, which often follow negative binomial or zero-inflated distributions [45].
- Batch effects: Differences in sequencing conditions, lab environments, or protocols lead to unwanted technical variation between datasets that can mask biological signals[11]

These challenges can be obstacles for researchers and scientists in single-cell experiments. To handle these disadvantages, simulation or synthetic single-cell is an ideal solution.

3.1.4 Generative modeling approaches in single-cell data

These synthetic dataset can be applied for many purposes:

- **Data augmentation for training classifiers or integration:** using synthetic single-cell data can enhance model robustness, especially in tasks like classification, clustering, or trajectory inference where real data may be scarce, noisy, or imbalanced across cell types or states. For example, classifiers trained on a combination of real and synthetic data often exhibit improved generalization to novel or rare cell populations [49].
- **Benchmarking and tool evaluation:** The field of single-cell genomics is rapidly evolving, with many tools being developed for dimensionality reduction, clustering, batch correction, or differential expression. Evaluating these tools on real datasets is limited by the lack of ground truth. Simulated data, with known cell types or trajectories, provide a controlled environment for performance comparison and reproducibility [27].
- **Ethical/technical constraints:** For example, access to patient-derived datasets, especially from clinical or pediatric populations, may be restricted due to privacy concerns or low sample availability. Synthetic data generation enables researchers to mimic such datasets while avoiding direct exposure to sensitive patient-level data

Moreover, synthetic data generation allows for controlled perturbation studies, such as simulating the effect of a genetic knockout or drug treatment. These *in silico* experiments are particularly useful in hypothesis generation and reducing experimental costs.

The generated dataset is often built based on the assumptions where the original dataset follows a pre-defined distribution, then a deep learning model is integrated to study the distribution parameters which capture the characteristics of the dataset, finally we can use these parameters to sampling new data. Therefore, we can evaluate the model by comparing the real data distribution and the generated data distribution, using statistical metrics, which are described in the following section.

3.2 Mathematical background

Probabilistic generative models rely on carefully chosen prior probability distributions to model the latent variables that underlie observed high-dimensional data. In the context of

single-cell RNA-seq (scRNA-seq) data, the choice of prior can strongly affect the learned representation's ability to capture biological variability and structure. This section introduces the mathematical formulations and biological relevance of three key prior distributions: Normal (Gaussian) Distribution, tDistribution and Lognormal Distribution, then compares their implications. The mathematical definition is referenced from [29].

3.2.1 Gaussian Distribution

The Gaussian distribution, also known as the normal distribution, is one of the most fundamental probability distributions in statistics and machine learning.

- Definition and probability density function (pdf): A random variable x is said to be normally distributed with mean μ and variance σ^2 , written as $x \sim \mathcal{N}(\mu, \sigma^2)$, if its probability density function (pdf) is given by:

$$f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

- Key properties: the Gaussian distribution has a symmetry PDF around the mean, an unimodal shape with a single peak at μ and decays exponentially at both ends. The mean represents the central tendency, and the variance reflects spread
- The multivariate Gaussian generalizes the above to d -dimensional space, $X \sim \mathcal{N}(\mu, \Sigma)$. The covariance matrix (Σ) defines the shape and orientation of the distribution where the PDF is defined by:

$$f(X) = \frac{1}{(2\pi)^{\frac{d}{2}} |\Sigma|^{\frac{1}{2}}} \exp\left(-\frac{1}{2}(X - \mu)^T \Sigma^{-1}(X - \mu)\right)$$

Where for a multivariate random vector:

$$X = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_m \end{bmatrix} \in \mathbb{R}^n$$

with mean vector:

$$\mu = \mathbb{E}[X] = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_n \end{bmatrix}$$

The covariance matrix $\Sigma \in \mathbb{R}^{n \times n}$ is defined as:

$$\Sigma = \mathbb{E}[(X - \mu)(X - \mu)^T]$$

- In single cell RNA context, the mean vector μ and the covariance matrix Σ , encoding for the variance and pairwise correlations between dimensions. The Gaussian prior is widely used in latent variable models such as VAEs and diffusion models due to (1) mathematical convenience: analytically simple, which can be fully described by just 2 parameters (μ, σ^2) , support with gradient based optimization, easier to sample from the Gaussian latent space during training than a skewed distributions and (2) well-behaved geometry: the latent space can have a symmetry, continuity, isotropy (uniform in all direction) shape, which allows easy training and result interpreting [17].

However, scRNA-seq data is often sparse, non-negative and heavily skewed, which may lead to misalignment between the Gaussian prior and the true data distribution in latent space. Some alternatives such as tdistribution or log-normal distribution can be experimented for more insights.

3.2.2 Student's T-Distribution

A random variable X is said to follow a Student's t-distribution with ν degrees of freedom, written as $X \sim t_\nu$, if the probability density function is given by:

$$f(X) = \frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\pi\nu}\Gamma(\frac{\nu}{2})} \left(1 + \frac{X^2}{\nu}\right)^{-\frac{(\nu+1)}{2}}$$

Where ν is the number of degrees of freedom, as the number of degrees of freedom grows, the t distribution approaches the normal distribution with mean 0 and variance 1. Γ is the gamma function, which is defined as $\Gamma(n) = (n-1)!$

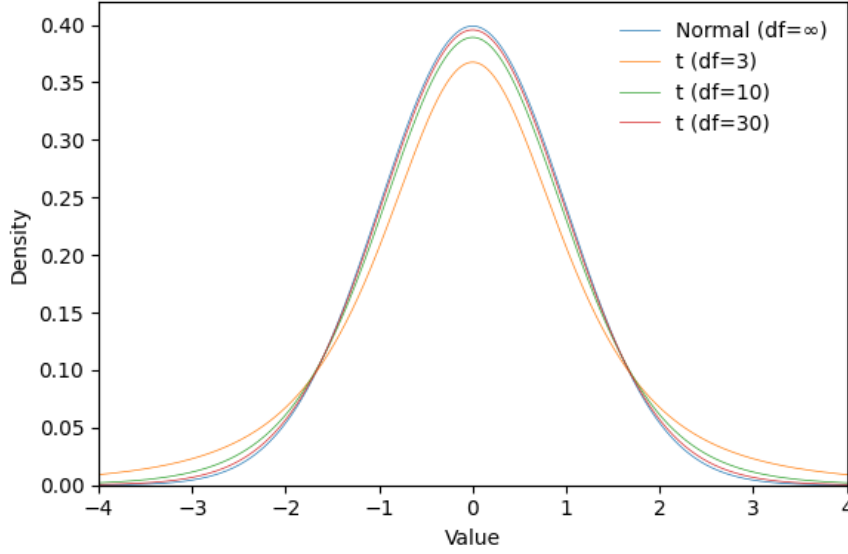


Fig. 3.4 Normal distribution and Student T-distribution with different parameters

Compared to the Gaussian, the Student's t has heavier tails, giving more probability mass in the tails—controlled by ν , which can be used to model outliers or skewed, sparse data.

3.2.3 Lognormal Distribution

The lognormal distribution is defined as the distribution of a random variable whose logarithm is normally distributed. If $x \sim \mathcal{N}(\mu, \sigma^2)$ then $x = \exp(y) \sim \text{LogNormal}(\mu, \sigma^2)$. The probability density function is defined as:

$$f(y) = \frac{1}{y\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(\log(y)-\mu)^2}{2\sigma^2}\right), y > 0$$

The distribution is strictly positive and skewed, with a heavier right tail, making it highly aligned with the observed distributions of gene expression in scRNA-seq data, which are (1) non-negative, since gene counts cannot be negative, (2) overdispersed, the variance often exceeds the mean and (3) right-skewed, fewer genes with high expression, many with low or zero expression

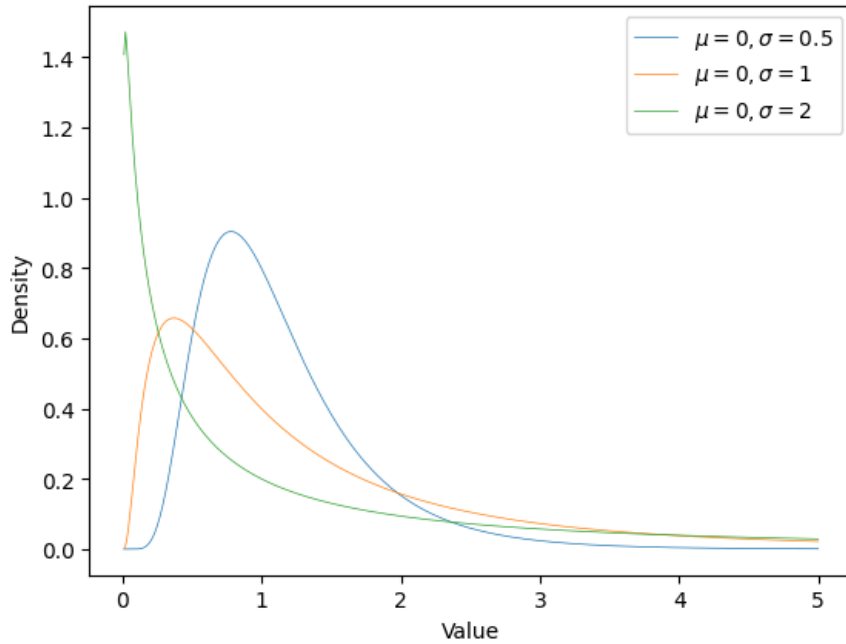


Fig. 3.5 Log-normal Distribution with different parameters

Due to its properties, Log-normal priors are biologically motivated and can potentially produce latent spaces and synthetic outputs that more closely reflect the statistical properties of real scRNA-seq data [42].

Comparison of probability distributions

1. **Impact on latent space geometry:** The geometry of the latent space defines how the generative model organizes and relates latent representations of the data. This affects how well the model captures biological variation and how interpretable the latent structure is.
 - Gaussian and Student's t prior creates a latent space that is symmetric, continuous, and centered around zero. The geometry is Euclidean, and distances between latent vectors are meaningful, which supports smooth interpolation between points. This is especially useful when modeling continuous cell state transitions, such as differentiation trajectories in developmental biology [17].
 - Lognormal prior, on the other hand, induces a skewed and multiplicative latent space. Since the lognormal is the exponential of a normal variable, it introduces nonlinearity and asymmetry, concentrating density near zero with a long tail. This results in a more irregular latent geometry, which may better match the distribution of raw gene expression levels, but can complicate learning and interpretation [45].

2. **Influence on Sampling Quality:** The sampling quality refers to how realistic and diverse the generated data is, which is especially important when synthetic single-cell data is used for augmentation, benchmarking, or hypothesis generation.

- Gaussian and Student’s *t* priors support smooth, balanced, and diverse sampling. Because the prior is symmetric and unimodal, randomly drawn latent codes are likely to land within the “valid” region of the latent space learned during training. This improves the validity and diversity of generated gene expression profiles.
- Lognormal priors, due to their heavy-tailed nature, can generate more extreme or rare-like samples, which may better reflect biological outliers or rare cell types. However, samples from the tails may be unrealistic if the model hasn’t seen enough such examples during training, leading to poor generalization or artifacts.

For example, a study by Lopez [22] on the scVI model noted that using a simple Gaussian prior with a sufficiently expressive decoder could still recover biological variability effectively. However, further works suggest that relaxing the Gaussian assumption or learning the prior from data could further improve sample realism and alignment with observed gene expression [39].

3.2.4 Statistical Distance Metrics

Statistical distance metrics are fundamental tools for quantifying how similar or different two probability distributions are. In the context of scRNA-seq generative modeling, these metrics help determine how well a model-generated dataset captures both the global and local structure of real single-cell data. Because scRNA-seq data are typically high-dimensional, sparse, and heterogeneous, the choice of metric critically influences both the evaluation and the interpretation of generative model performance. The definitions are referenced from [33].

1. **Energy Distance (ED)** Energy Distance is a non-parametric statistical measure of dissimilarity between two probability distributions based on the distances between random samples from these distributions. Given two samples $X = \{x_1, x_2, \dots, x_n\}$ from distribution P and $Y = \{y_1, y_2, \dots, y_m\}$ from distribution Q , the squared energy distance is:

$$D_{ED}^2(P, Q) = 2E\|X - Y\| - E\|X - X'\| - E\|Y - Y'\|$$

Where $X, X' \sim P$ and $Y, Y' \sim Q$ are i.i.d random variables and $\|\cdot\|$ is the Euclidean norm

- The distance is equal to zero if and only if X and Y follow the same distribution.
- ED is particularly suitable for high-dimensional data, such as gene expression profiles, since it does not rely on parametric assumptions and captures both global and marginal distributional differences. This is important in scRNA-seq data, which often exhibits multimodal gene expression across cell types. Furthermore, energy distance metric has a low computational cost with $O(n^2)$ in the sample size and the differentiability properties, which is compatible for integrating into neural networks penalty components to promote generalization ability and avoid mode collapse.

2. Maximum Mean Discrepancy (MMD)

Maximum Mean Discrepancy is a non-parametric, kernel-based statistical distance measuring how far apart two probability distributions are in a reproducing kernel Hilbert space (RKHS). Given two samples $X = \{x_1, x_2, \dots, x_n\}$ from distribution P and $Y = \{y_1, y_2, \dots, y_m\}$ from distribution Q , the (squared) MMD under kernel $k(\cdot, \cdot)$ is:

$$MMD_k^2(P, Q) = 2E_{P, Q}[k(X, Y)] - E_P[k(X, X')] - E_Q[k(Y, Y')]$$

Where $X, X' \sim P$ and $Y, Y' \sim Q$ are i.i.d random variables and k is any positive-definite kernel (e.g. Gaussian or Laplacian), $k(X, Y) = (\phi(X), \phi(Y))$ which maps 2 points X, Y to a kernel space for distance comparison.

- MMD quantifies the distance between 2 probability distributions by comparing their mean embeddings (the representations of these distributions in a higher-dimensional feature space). The value for MMD will be zero if the two distributions are the same and will increase when the distributions diverge.
- One feature of MMD is the kernel function, which is used to map data points into a higher-dimensional space, where the distributions can be more easily compared. This function computes the similarity (inner product) between data points.

The formula is coming from the mathematical basis described below.

Given a probability measure P on X , feature means (mean embedding) is a feature map that takes $\phi(X)$ and maps it to the means of every coordinate of $\phi(X)$:

$$\mu_P(\phi(X)) = [E[\phi(X_1)], \dots, E[\phi(X_m)]]^T$$

The inner product of feature means of $X \sim P$ and $Y \sim Q$ can be written in terms of kernel function such that:

$$\langle (\mu_P(\phi(X)), \mu_Q(\phi(Y))) \rangle = E_{P,Q}[\langle \phi(X), \phi(Y) \rangle] = E_{P,Q}[k(X, Y)]$$

Given X, Y , the maximum mean discrepancy is the distance between feature means of X, Y :

$$MMD^2(P, Q) = \|\mu_Q - \mu_P\|^2$$

If we use the norm induced by the inner product such that $\|x\| = \sqrt{\langle x, x \rangle}$, the equation above becomes:

$$MMD^2(P, Q) = \langle \mu_P - \mu_Q, \mu_P - \mu_Q \rangle = \langle \mu_P, \mu_P \rangle - 2\langle \mu_P, \mu_Q \rangle + \langle \mu_Q, \mu_Q \rangle$$

Then using the kernel function, we can derive the formula for MMD as mentioned above.

3. Spearman's Rank Correlation Coefficient(Spearman SCC)

Spearman's Rank Correlation (SCC) is a non-parametric measure of monotonic relationship (consistently increasing or decreasing) between two variables. Given a paired observations (X_i, Y_i) ranked as $R(X_i)$ and $R(Y_i)$, in which the rank of an observation is its position in a sorted sequence of values, the Spearman correlation coefficient ρ is defined as:

$$\rho = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)}$$

where $d_i = R(X_i) - R(Y_i)$ is the difference between ranks and n is the number of observations

- Spearman's SCC range between -1 and 1 , in which 1 indicates a perfect positive monotonic relationship, 0 is no monotonic correlation and -1 for a perfect negative monotonic relationship.
- SCC is robust to outliers, less sensitive to data distribution assumptions and does not assume normality or linearity relationships, making it suitable for evaluating relationships in high-dimensional and noisy datasets such as scRNA-seq data, where the relationships between real and synthetic gene expressions are more meaningful than exact numeric correspondence

For the ideal case, a $\rho \approx 1$ indicating a strong positive monotonic correlation between synthetic and real data distributions. Typically $\rho \geq 0.7$ are considered good, reflecting a substantial monotonic relationship, which indicates that synthetic data adequately preserves the ranking structure present in real data. A value close to 0 or negative meaning that a weak or inverse rank relations between synthetic and real data.

3.3 Machine Learning and Deep Learning background

3.3.1 Overview of application in Biological Data Analysis

- Machine Learning (ML) refers to algorithms that learn patterns from data. ML approaches can be categorized into:
 - Supervised learning: where labels guide learning (e.g., predicting cell type).
 - Unsupervised learning: where structure is learned without labels (e.g., clustering, dimensionality reduction).

In genomics and single-cell data, ML methods are used for clustering cells, classifying cell types, detecting rare populations, and reducing dimensionality (e.g., PCA, t-SNE, UMAP) [18]. With the development of Machine Learning and Deep Learning algorithm, the applications can be found at several sectors, one of them is using Generative algorithm to modeling for biological data, which will be describe in the next section.

3.3.2 Generative Modeling Concepts

Generative models aim to learn the underlying distribution of observed data to produce synthetic samples that resemble real data. In the context of scRNA-seq, generative models can address challenges such as high dimensionality, sparsity, and batch effects while generating synthetic cells for data augmentation, benchmarking, and interpretability. This section will provide the fundamental concept about generative model that will be further used for the thesis.

Generative models can generate new data instances; in contrast, discriminative models differentiate between different kinds of data instances. For example, a generative model could create new scRNA-seq gene expression matrix which is similar to real matrix, while a discriminative model could tell the difference between cell types. Formally, given a set of data instances X and a set of labels Y :

- Generative models: capture the joint probability $P(X, Y)$ or just $P(X)$ if there are no labels.
- Discriminative models: capture the conditional probability $P(Y|X)$.

A generative model includes the distribution of the data itself, then tells how likely a given example is, while a discriminative model will ignore the question about whether a given instance is likely, and just gives information about how likely a label is to apply to the instance.

3.3.3 Deep Generative Models

Generative models have emerged as powerful tools to address the challenges of scRNA-seq data by learning the underlying data distribution and generating new, realistic cell profiles. Three main categories of generative models are used in single-cell biology: Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs), and more recently, Diffusion Models. For explanation purpose, we used example as image generation model but the overall model architecture and intuition stay the same.

1. **Variational Autoencoders (VAEs):** VAEs are a class of deep generative models that learn a probabilistic latent space using an encoder-decoder architecture [17]. The encoder maps data X into a distribution over latent variables $P_\phi(Z||X)$, while the decoder reconstructs x from samples of Z using $P_\theta(\hat{X}||Z)$.

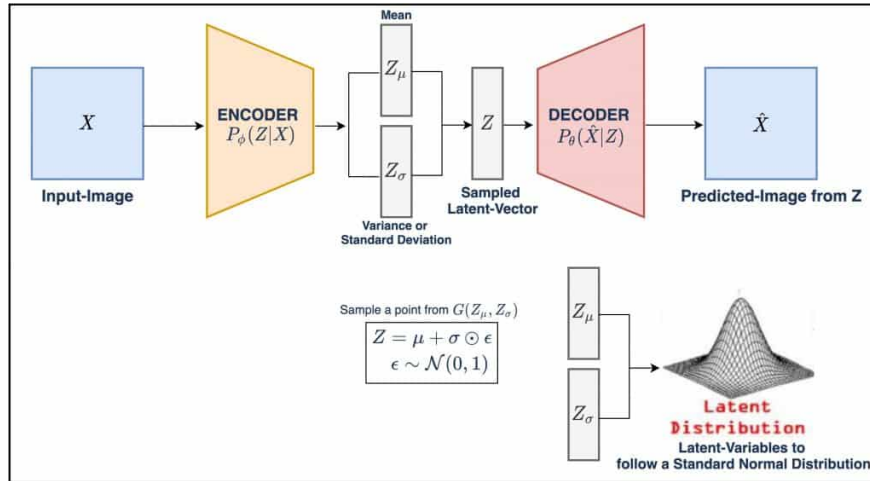


Fig. 3.6 Variational Autoencoders (VAEs) architecture [36]

A key aspect of VAEs is the use of a prior distribution $p(z)$ —commonly a Gaussian—which defines the geometry of the latent space. This prior plays a crucial role in guiding the model toward smooth, disentangled representations that generalize well to unseen cells [22]

2. **Generative Adversarial Networks (GANs):** GANs consist of two neural networks: a generator $G(z)$, which learns to produce synthetic data from latent variables, and a discriminator $D(x)$, which learns to distinguish between real and fake samples [8]. The networks play a minimax game: the generator tries to fool the discriminator, and the discriminator tries to avoid being fooled. Both the generator and discriminator are neural networks, the generator output is connected directly to the discriminator input.

Using backpropagation, the discriminator's classification provides a signal that the generator uses to update its weights. Despite their success in image synthesis, GANs

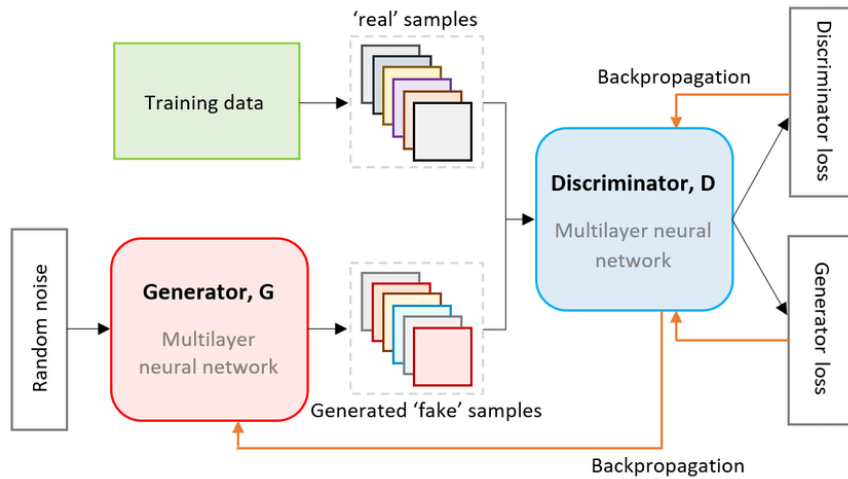


Fig. 3.7 Generative Adversarial Networks (GANs) architecture [21]

are less stable in training, prone to mode collapse, and often struggle with discrete, sparse data like scRNA-seq [27]

3. **Diffusion Models:** This is a class of generative model based on iteratively adding noise to data (forward process) and learning to reverse this process to generate new samples (reverse process) [13]. These models have shown superior sample quality and stable training compared to GANs, especially for high-dimensional and multimodal distributions like scRNA-seq. They are robust to noise and can capture complex biological processes more accurately [25].

Diffusion Model is the fundamental building block for Stable Diffusion Model which will be explained in the next section.

3.3.4 Stable Diffusion Models for single-cell data

Stable diffusion models are a class of deep generative models based on a two-step noising and denoising process that learns the data distribution by modeling the reverse of a diffusion (noise) process. Formally, these models define a Markov chain that gradually adds Gaussian noise to the data in a sequence of steps (forward process), and then train a neural network to reverse this process and recover the original data (reverse process) [13].

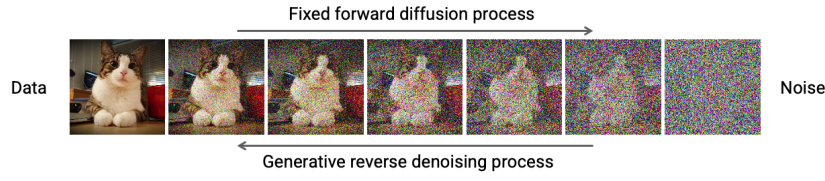


Fig. 3.8 Diffusion model architecture [44]

The stable diffusion model is based on the original Denoising Diffusion Probabilistic Models (DDPMs). The architecture is described in the following section.

3.3.4.1 Overview of stable diffusion models

Stable Diffusion, or Latent Diffusion Models (LMD) developed by Rombach et al. in 2022 [34] introduces a modification by performing the diffusion process in the latent space. This can be achieved by training an encoder E for encoding a fullsize image to a lower dimension representation (latent space). Then doing the forward diffusion process and reverse process within this latent space. Lastly, with a trained decoder D , we can decode the image from the latent representation back to the pixel-space (image original space).

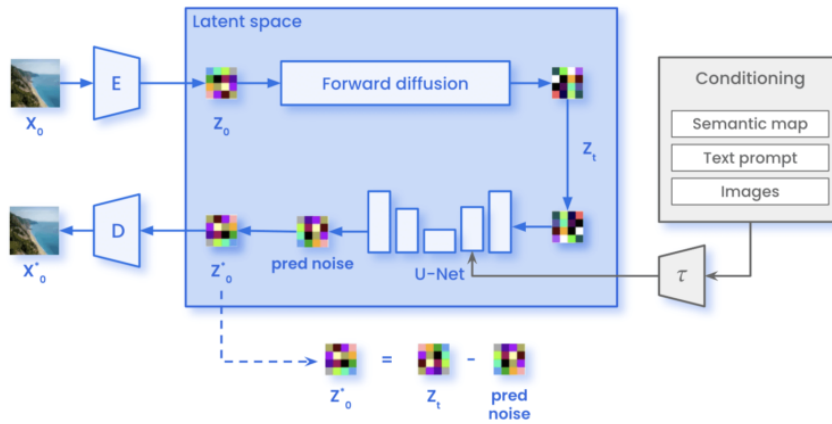


Fig. 3.9 Stable Diffusion model architecture [4]

Formally, during training, the original data X_0 are encoded using encoder E , gives the latent representation of the original data Z_0 . In the forward diffusion, the image is destroyed by adding Gaussian noise, return a noisy data Z_T . This noised is then passed through the reversed diffusion process, in order to predict the noise in Z_T using a neural network (originally U-NET). The generated image is created by subtract Z_T with predicted noise and

update through backpropagation until converged. Conditions τ such as text prompts and images can be added during this process.

For scRNA-seq gene expression matrix, the use of encoder-decoder pairs may not be necessary, since the main purpose of using Variational Autoencoders (VAE) is to compress high-res images into a lower-dimensional latent space. For a flat, tabular gene-expression matrix with less than 1000 dimensions, we already have a relatively compact vector representation, the use of VAE can introduce unstable and complexity for the model to learn [34].

The above idea has been developed into a training algorithm for diffusion model which is described below.

3.3.4.2 Training process of stable diffusion model

Diffusion model containing 2 main processes: Forward diffusion and Reverse diffusion. During the forward process, the model destroys the original data by adding noise to it. These noises added will follow a noise scheduler to specify the amount of noise for each time steps. The reversed process will try to reconstructing the original data base on the noise by using a deep neural network to learn to predict the noise. After training, new data are generated by sampling random noise and passing through the learned reversed diffusion process.

The mathematical explanation is referenced from [7] and [14]

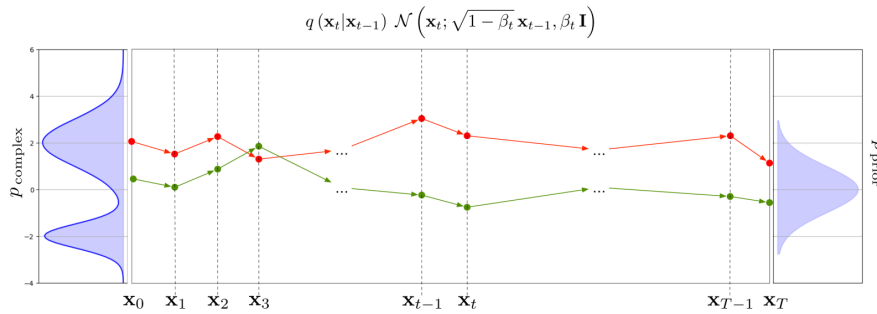


Fig. 3.10 Diffusion process with probability distribution view [14]

The main idea of the diffusion model is using 2 Markov chains, a forward to noise data and a reversed to convert noise back to data. A Markov chain represents a sequence of continuous states, where the next state depends only on the current state and the transitions between states occur according to a defined probability distribution. This would help simplify the calculation and enable the feasibility of further calculations, especially using the variational inference technique to approximate complex probability distributions that are computationally intractable, which appear in the reversed diffusion process.

Notation:

- X_0 is the original data sample. A sequence of latent variables can be expressed as $X_0, X_1, \dots, X_T$ with the same dimension as X_0
 - q is the data distribution with $q(X_0)$ is the real data distribution of sample X_0
 - β is the diffusion rate (variance of the distribution), define by a noise scheduler for each β_t at timestep t .
1. **Forward diffusion:** this is a step-wise transformation of a sample data X_0 to a noisy data X_T . During this process, the Gaussian noise is incrementally added at each iteration. This can be represented as a Markov Chain as:

$$q(X_t|X_{t-1}) = \mathcal{N}(X_t; \sqrt{1 - \beta_t}X_{t-1}, \beta_t I)$$

where $\mathcal{N}$ is Gaussian distribution with 2 parameters, mean $\mu = \sqrt{1 - \beta_t}X_{t-1}$ and covariance matrix $\Sigma = \beta_t I$, with I is the identity matrix.

The joint probability of the entire process from the original data sample X_0 to the final sample at timestep T can be written as the product of all previous Markov Chain transitions:

$$q(X_{1:T}|X_0) = \prod_{t=1}^T q(X_t|X_{t-1})$$

At timestep T (last timestep): mean and variance are close to 1 and 0, mathematically speaking, $q(X_T) \approx p_{prior}(X_T) = \mathcal{N}(X_T; 0, I)$.

Instead of creating an algorithm to add noise into original data repeatedly, which is computationally expensive, we can have a closed-form formula to directly sample a noisy image at time step t by the reparameterization trick. The full formula derivation can be found in the original paper [13]. For the scope of this thesis, we mention the final formula only.

$$X_t = \sqrt{\bar{\alpha}_t}X_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon$$

with $\epsilon \sim \mathcal{N}(0, 1)$ and $\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$ is the cumulative product of $\alpha_t = 1 - \beta$

We do not need to train any machine learning model for this forward process, but we will follow a noise-adding mechanism.

2. **Noise scheduling:** There are 2 common types, linear and cosine scheduler [7].

- **Linear schedule:** using this strategy, the noise is added or removed at a constant rate throughout the process. The variance β_t increases uniformly from an initial value β_0 at the first timestep to a final value β_T at the last timestep. The same method is applied for reverse process, the noise level will decrease linearly. With any $t \in [1, T]$, the noise follows a predefined linear progression:

$$\beta_t = \beta_1 + \frac{t-1}{T-1}(\beta_T - \beta_1)$$

- **Cosine schedule:** compare to linear noise schedule which is straightforward for implementation but lacks of flexibility and adaptive, a cosine noise schedule gives a gradual and adaptive noise progression by following a trajectory similar to cosine function. With $\beta_t = 1 - \frac{\alpha_t}{\alpha_T - 1}$, which is the relationship between β_t and α to ensure the rate of change in the noise schedule is highest when α is at its mid-point, we can define the cosine noise schedule in terms of α as:

$$\alpha_t = \frac{f(t)}{f(0)}, f(t) = \cos\left(\frac{t+s}{1+s} \frac{\pi}{2}\right)^2$$

where s is a small constant added to avoid β_t approach zero at the beginning of the timestep.

At timestep 0 or T , the rate of change smooths out, meaning that only a minimal amount of noise is added to the data distribution in these regions.



Fig. 3.11 Linear (top) and cosine (bottom) noise scheduler [30]

The difference between these two schedulers is their denoising complexity. The cosine scheduler gives more challenging denoising tasks until after the midpoint of training, which can improve sample quality and enhance training efficiency.

3. **Time Embedding:** During denoising, the model incorporates the current time step t via positional/time embeddings, allowing the neural network to condition its predictions on the stage of the diffusion process [30].
4. **Reversed diffusion:** during this process, the aim is to remove noise and reconstruct the original data sample. We will train a deep learning model to approximate the noise and learn to remove it from a Gaussian-distributed latent variable. The model is trained by minimized a loss function, which is a variational bound with negative log-likelihood.

Originally, we want to calculate directly the reversed distribution using Bayesian:

$$q(x_{t-1}|x_t) = q(x_t|x_{t-1}) \frac{q(x_{t-1})}{q(x_t)}$$

but $q(x_t) = \int q(x_t|x_{t-1})q(x_{t-1})dx$ need the whole data distribution (of the predicted data) $q(x_{t-1}) \rightarrow$ intractable.

Therefore, we can define a new distribution that represent the reverse process directly: $p_\theta(x_{t-1}|x_t)$ and assume that for small enough forward steps, i.e. $\beta_t \in (0, 1)_{t=1}^T$, the reverse process step $q(x_{t-1}|x_t)$ can also be estimated as a Gaussian distribution, therefore, we can parametrize this process as:

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \sigma_\theta(x_t, t))$$

and using a neural network to learn these parameters θ .

Since the reverse process is also a Markov chain, the joint probability for a sequence of samples is:

$$p_\theta(x_{0:T}) = p(x_T) \prod_{t=1}^T p_\theta(x_{t-1}|x_t)$$

We then want to maximize log likelihood of observed data, since the likelihood $p_\theta(x_0)$ quantifies how probable the observed data x_0 is under the model. Maximizing this is equal to finding the best estimate distribution. After a complex process of constructing the Evidence Lower Bound (ELBO), decompose it into many KL-divergences (which are use to measure the similarity between two data distributions), then reformulate these KL which is proved in the original paper [13] our goal becomes a noise-prediction task and minimize the loss function of the form:

$$L(\theta) := \mathbb{E}_{t, x_0, \theta} [||\epsilon - \epsilon_\theta(x_t, t)||^2]$$

5. **Model training:** the training process can be summarized in below algorithm.

Algorithm 1 Training reversed process1: **repeat**2: $x_0 \sim q(x_0)$ 3: $t \sim \text{Uniform}(\{1, \dots, T\})$ 4: $\varepsilon \sim \mathcal{N}(0, I)$

5: Form noisy input:

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \varepsilon$$

6: Take gradient step on

$$\nabla_{\theta} \|\varepsilon - \varepsilon_{\theta}(x_t, t)\|^2$$

7: **until** converged

6. **Data generation:** to sampling data for a specific timestep, we can use the sampling algorithm defined as below.

Algorithm 2 Sampling data

1: Draw initial noise:

$$x_T \sim \mathcal{N}(0, I)$$

2: **for** $t = T, T - 1, \dots, 1$ **do**3: **if** $t > 1$ **then**4: $z \sim \mathcal{N}(0, I)$ 5: **else**6: $z \leftarrow 0$ 7: **end if**

8: Compute preconditioned denoised step:

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \varepsilon_{\theta}(x_t, t) \right) + \sigma_t z$$

9: **end for**10: **return** x_0 **3.3.4.3 Diffusion model in scRNA-seq data**

As mentioned above, scRNA-seq data pose several modeling challenges, including high-dimensionality, sparsity and multimodality. The stable diffusion models are excellent candidates to address these challenges due to:

- Robustness: Diffusion models are robust to noise and data corruption, which is ideal for imputation and denoising.
- Modeling ability: The capacity to model complex, multimodal distributions while preserving cell type diversity is also a valuable point to adapt this method into single cell data.
- Ability to generate high-fidelity synthetic cells, which match both global and local structures in gene expression manifold [25]

In comparison with VAEs, which often generate blurry or oversmoothed samples, or GANs, which suffer from mode collapse, diffusion models maintain the granularity of gene expression heterogeneity, a crucial feature for biological interpretability.

3.3.4.4 Conditional Diffusion Model for scRNA-seq data

One of the key requirements for data generation is the conditional data synthesis based on user-specified requirements. In the context of image generation, this condition can be a text-prompt or an image [51]. In scRNA-seq data generation field, this conditional can be a cell type classifier, this method does not integrate in the two forward and reversed processes but is trained separately instead by using condition labels, cell types for instance and provides gradients to guide cell generation. There has been a variety of deep-learning based cell type classifiers which can perform this gradients generation task and guide the conditional generation process, such as scANVI [23], MARS [5], and scFoundation [12]. Another approach is to do direct conditional modeling by integrating conditions directly into the diffusion process, which is intuitive and simple for both training and inference procedures [51]. We can train the usual denoising loss as:

$$L(\theta) := \mathbb{E}_{t, X_0, \theta, y} [||\varepsilon - \varepsilon_{\theta}(x_t, t, y)||^2]$$

with y is the condition that has been embedding (one-hot, text encoder,...), in this case will be the cell types.

3.3.5 Evaluation of Generative Models

Evaluating generative models is essential to ensure that the synthetic data generated not only looks realistic but also captures the true underlying distribution and biological variability of the data. This is especially important in single-cell omics, where distributional accuracy, gene expression fidelity, and cell-type heterogeneity are critical for downstream biological

analysis. This section outlines common quantitative metrics and model validation strategies used to assess how well synthetic data generated by deep generative models mimics real scRNA-seq data.

To compare the similarity between generated and real cells, we can evaluate the generated data with several statistical metrics, for instance, Spearman Correlation Coefficient (SCC), Maximum Mean Discrepancy (MMD), and quantile-quantile plot (QQ-plot).

The nonstatistical metrics that can be used are Uniform Manifold Approximation and Projection (UMAP) visualization, marker gene expression, CellTypist classification, random forest evaluation, and KNN evaluation. The UMAP plot was used to visualize the generated and real expression data on a 2D plane to provide a subjective judgment of the generated data.

In the scope of this thesis, we adopt 3 statistical metrics which are Energy Distance, MMD and SCC described in section 2.2.4, as well as the nonstatistical metrics including UMAP visualization and KNN AUC score. These metrics can be used to reflect how well the model preserves gene expression profiles, which is crucial for biological interpretability.

3.3.5.1 Uniform Manifold Approximation and Projection (UMAP)

Uniform Manifold Approximation and Projection (UMAP) is a dimensionality reduction and visualization method widely used to qualitatively evaluate the structure of high-dimensional datasets. Unlike statistical metrics, UMAP focuses on preserving local and global data structures visually, thus aiding in intuitive evaluation of data quality and similarity.

Given a high-dimensional dataset $X = \{x_1, x_2, \dots, x_n\}$ where $x_i \in \mathcal{R}^D$, UMAP maps these points to a low-dimensional representation $Y = \{y_1, y_2, \dots, y_n\}$ with $y_i \in \mathcal{R}^d$, usually $d = 2$ for visualization purposes. UMAP aims to approximate and preserve the topological structure of the data manifold from the original high-dimensional space. The embedding Y is derived through minimizing the following cross-entropy based objective function, balancing local connectivity and global relationships:

$$C = \sum_{i \neq j} [\mu_{ij} \log \frac{\mu_{ij}}{v_{ij}} + (1 - \mu_{ij}) \log \frac{1 - \mu_{ij}}{1 - v_{ij}}]$$

where μ_{ij} represents the probability that points i and j are neighbors in the high-dimensional space and v_{ij} represents the probability of being neighbors in the low-dimensional embedding.

- UMAP helps intuitively identify the quality of the generated data without relying on numeric thresholds or formal statistical criteria.
- The points clustering closely in UMAP plots indicate high similarity or structural closeness in original high-dimensional space.

- A clear overlap between synthetic and real datasets on UMAP visualizations implies that synthetic data effectively captures and reproduces original data structure. In contrast, the distinct separation or divergence between the two datasets indicate shortcomings in data generation.

3.3.5.2 K-Nearest Neighbors Area Under Curve (KNN-AUC)

This is a metric that evaluates how effectively generated data points can preserve the local neighborhood structure observed in the original data distribution. It assesses the ability of synthetic samples to reproduce realistic local structures. Given two sets of points $X = x_1, x_2, \dots, x_n$ from real distribution P and $Y = y_1, y_2, \dots, y_n$ from synthetic distribution Q , in a binary classification problem distinguishing real with synthetic points, the KNN-AUC score measures how well a KNN classifier distinguishes real data points from synthetic ones. We construct a combined dataset $Z = X_{real} \cup X_{synthetic}$ with labels as $Labels = \{ 1 \text{ for real points, } 0 \text{ for synthetic points } \}$, then using a KNN classifier. For each point $z_i \in Z$, compute the fraction of its K-nearest neighbors belong to the real dataset. The score is computed as:

$$s(z_i) = \frac{1}{K} \sum_{z_j \in N_K(z_i)} I(z_j \text{ is real})$$

where $N_K(z_i)$ is the set of K-nearest neighbors to point z_i and $I(\cdot)$ is the indicator function: $I = 1$ if its argument is true, which means z_j is real and $I = 0$ otherwise.

These scores are then used to compute the ROC curve by varying thresholds between 0 and 1, for each threshold T :

- Predict label = real (1) if $s(z_i) \geq T$
- Predict label = synthetic (0) otherwise

We have the True Positive Rate (TPR) and False Positive Rate (FPR) at each threshold:

$$TPR = \frac{\text{True Positives}}{\text{Total Real Data}} \text{ and } FPR = \frac{\text{False Positives}}{\text{Total Synthetic Data}}$$

We then plot this and calculate Area Under the ROC Curve (AUC). The formula $KNN - AUC$ score is formally defined as:

$$KNN - AUC = AUC(labels = \{z_i\}, s(z_i))$$

- A $KNN - AUC$ close to 0.5 indicates that real and synthetic points are indistinguishable based on local neighborhood structures, reflecting high-quality synthetic data.
- This metric is informative for high-dimensional datasets such as scRNA-seq data because it directly captures the local geometry preservation, essential for biological interpretations.

Chapter 4

Material and Methodology

4.1 Material

4.1.1 Data source

The data used for this project is obtained from The Tabular Muris Consortium [35], provided a comprehensive scRNA-seq benchmark dataset comprising more than 100,000 cells from 20 organs and tissues, which ensures diversity and biological relevance. The data are stored in *.h5ad* format and are described as the following figure.

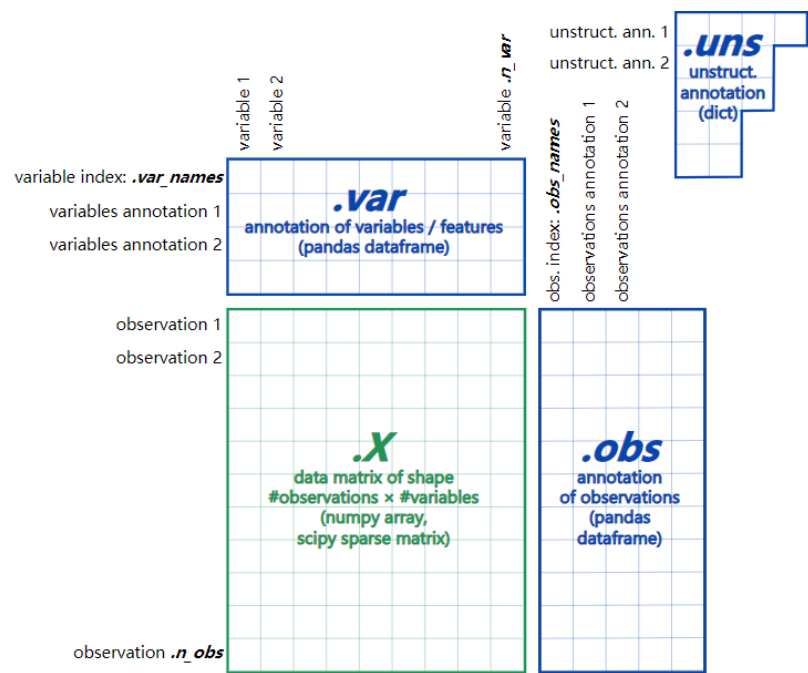


Fig. 4.1 Tabular Muris Format [2]

AnnData uses some terms to describe cells and genes, which are observations and variables, respectively. The four areas we can stored information are:

- *AnnData.X* for the count matrix (gene expression matrix)
- *AnnData.obs* for the metadata about the observations (cells)
- *AnnData.var* for the metadata about the variables (genes)
- *AnnData.uns* stores additional, unstructured information

For this thesis, we adapt a part of the dataset, with the 10 most common cells types and the description presented in the table below.

Cell Type	Number of samples
B cell	5918
Basal cell of epidermis	3964
T cell	3643
Keratinocyte	3536
Mesenchymal progenitor cell	1676
Fibroblast	1666
Chondrocyte	1630
Mesenchymal stem cell	1468
Endothelial cell	1418
Macrophage	1290
Total	45602

Table 4.1 Number of sample per cell type

The data are represented in the form of sparse matrix, with **45602 cells** (sample) and **128 genes** (feature). The entries values ranging from 0 to 9.4 and mostly 0 value since most of the genes are not expressed.

4.1.2 Software Framework and Environment

The thesis is implemented using Python programming language with the core libraries:

- Scanpy [47] - for preprocessing and exploratory data analysis of scRNA-seq data
- PyTorch [31]- for model implementation and training

- scikit-learn - for model evaluation and preprocessing tools
- NumPy, Pandas - for data manipulation
- Seaborn/Matplotlib/Joypy - for visualization

Hardware: the model are training on NVIDIA H100 SXM and L40S GPU.

4.2 Methodology

In this section, we present the overall pipeline to implement this project and the detailed explanation for each step in the process.

4.2.1 Pipeline

Our pipeline consists of 4 main stages: Data preprocessing, Model training, Data generation and Model evaluation, which are illustrated in the pipeline below.

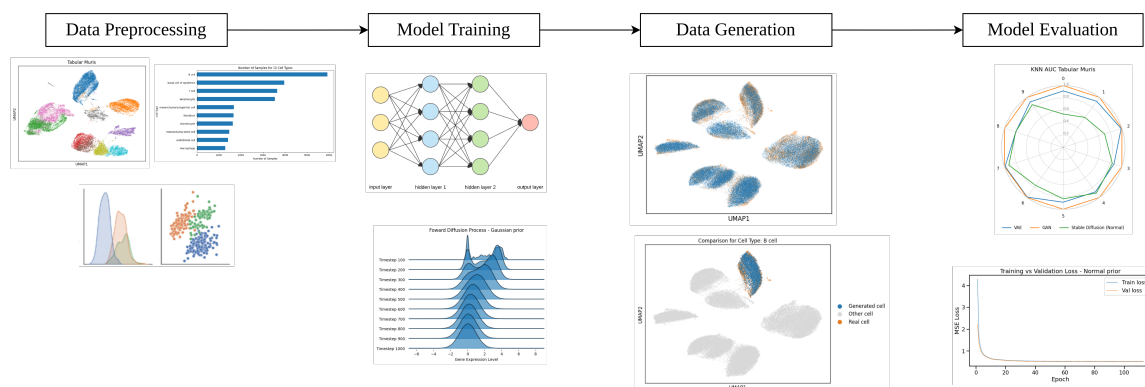


Fig. 4.2 Project Overall Pipeline

The methodology applied for each steps in the pipeline will be explained clearly in the following sections.

4.2.2 Data preprocessing

For data preprocessing and analysis, we adopt a part from the guideline provided by [3]

Quality control A key initial step in scRNA-seq analysis is identifying true cells by filtering out barcodes that likely correspond to empty droplets or background noise, often using UMI thresholds or methods like EmptyDrops. This is followed by quality control to

remove damaged or dying cells based on metrics such as gene count, mitochondrial RNA proportion, and mapping quality, with thresholds tailored to each dataset. Additionally, it's important to detect and remove doublets—barcodes capturing more than one cell—or barcode multiplets, where a single cell is labeled by multiple barcodes, using tools like Scrublet or DoubletFinder, though no method is universally accurate.

For Tabular Muris dataset, the quality control (QC) process has been adapted and the batch effects were shown that it is not the main source for variance in the dataset, so we can continue to do further analysis without methods of QC and batch correction.

Data Normalization The original gene expression values have been log-normalization into range $[0, 9.4]$. This normalization was necessary to correct for variability in sequencing depth, a pronounced issue in scRNA-seq data due to significant biological differences among cells, such as distinct cell-cycle stages, and technical variations like differing droplet sizes.

Feature Selection Due to the size of features (128 genes), no further method of feature selection is needed for this dataset. With a larger dimensions, a feature selection methods for instance PCA or Random Forest should be adopted to avoid redundancy and improve computational efficiency. The visualization of the dataset is described below.

Data Visualization After doing data preprocessing steps, we used UMAP for illustrate the dataset and qualitatively evaluate the structure of the original dataset. UMAP helps reduce high-dimensional data into a two-dimensional embedding, which captures the local and global data structure in a visual format.

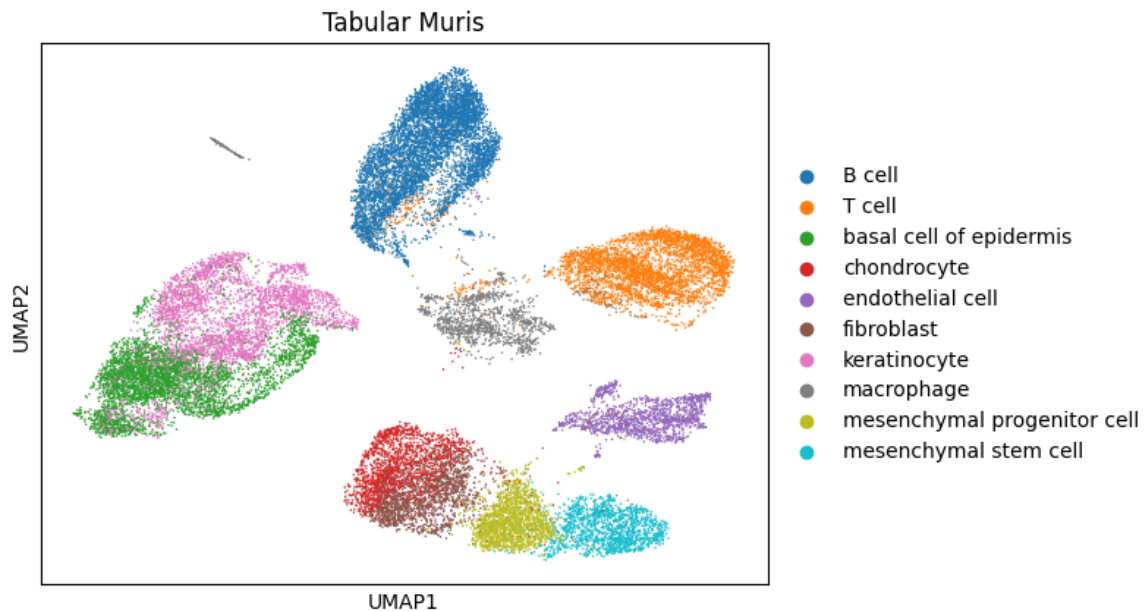


Fig. 4.3 UMAP visualization for 10 most common cell types

From the UMAP plot above, we can observed prominent clusters, such as B cells, T cells, and mesenchymal progenitor cells, appear well-separated, indicating strong differences in gene expression profiles among these types. Keratinocyte and basal cells of epidermis partially overlap, indicating similar gene expression patterns or close biological relationships.

Based on this biological structure of the dataset, we built a Stable Diffusion model with different prior settings to evaluate their effects on the data generation quality. The model implementation will be described in the next section.

4.2.3 Model configuration and training

4.2.3.1 Stable Diffusion Model

For the scope of this thesis, we implemented a stable diffusion model, with some modifications suitable for scRNA-seq data. The detailed architecture is illustrated below. For each process, we provided and overview about the input/output of the process and the architecture behind them.

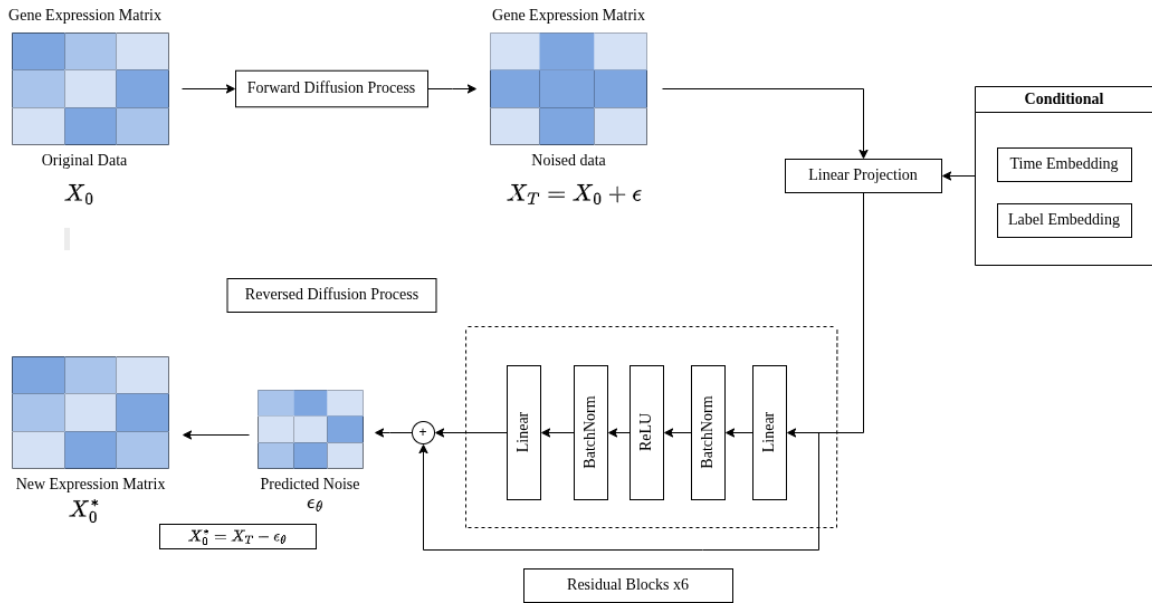


Fig. 4.4 Overall architecture of Stable Diffusion for scRNA-seq data

The model consists of 2 main processes:

1. **Forward Diffusion Process:** The forward process involves gradually adding noise to a clean data sample X_0 over multiple timesteps to produce a noised version X_T , which follows a prior distribution (typically Gaussian).

$$X_T = X_0 + \varepsilon \text{ with } \varepsilon \sim \mathcal{N}(0, 1)$$

In our implementation, three different diffusion processes for each prior assumptions are available:

- Gaussian prior (normal distribution): implemented as *DiffusionProcess*.
- Log-normal prior: implemented as *DiffusionProcessLogNormal*.
- Student T prior: implemented as *DiffusionProcessStudentT*

The process of adding noise and destroyed the original data structure can be visualized as below.

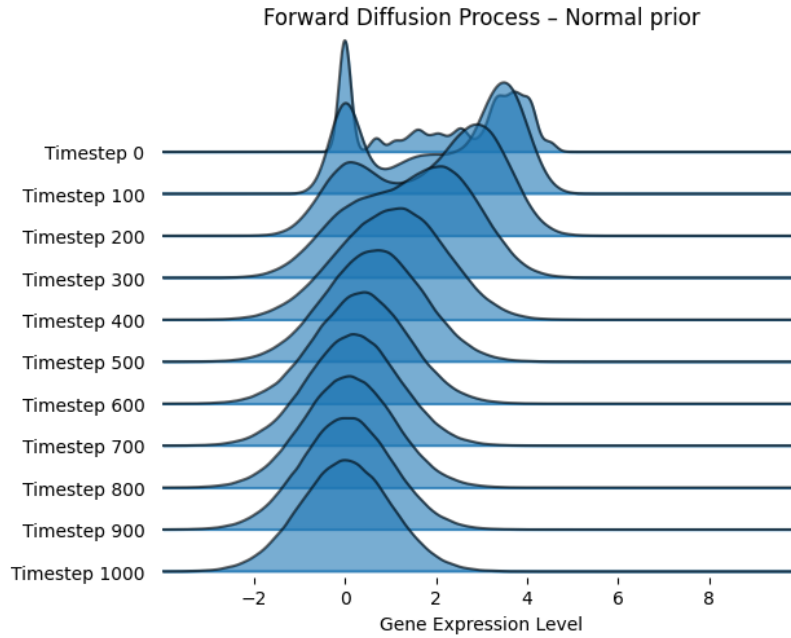


Fig. 4.5 Forward Diffusion process (Normal distribution noise)

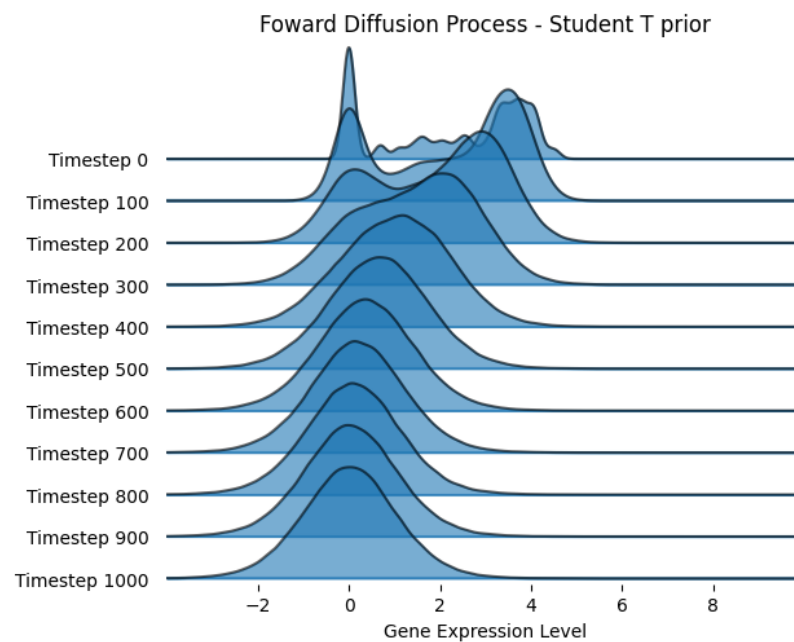


Fig. 4.6 Forward Diffusion process (Student's t distribution noise)

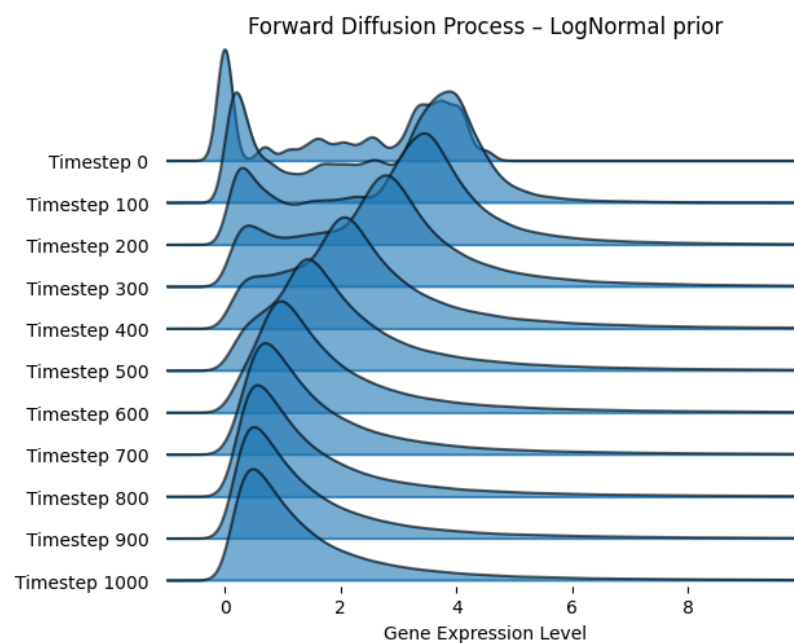


Fig. 4.7 Forward Diffusion process (Log-normal distribution noise)

The original data follow a distribution, it then gradually become another distribution, which follows our prior settings through timesteps.

2. **Reversed Diffusion Process:** The reverse process is learned by training a neural network to predict the added noise ε_θ at each timestep. The original data is calculated by subtract the noised data with predicted noise.

$$X_0^* = X_T - \varepsilon_\theta$$

In our model, the reverse diffusion is implemented as follows:

- Inputs: Combination of noisy sample X_T , time embedding t , label embedding.
- Architecture: The combined input is passed through a Linear projection layer to match the model's hidden dimension. The core of the model is a stack of Residual blocks:
 - Each block includes: Linear layer $\rightarrow$ BatchNorm $\rightarrow$ ReLU $\rightarrow$ BatchNorm $\rightarrow$ Linear
 - Skip connection: which adds the input of the block directly to the output and allows the model to preserve important input information and only learn the residual (difference/noise) needed to reconstruct the original data.

$$\text{Output} = x + F(x)$$

- Output: predicted noise ε_θ
- This process is repeated (from $T \rightarrow 0$) during sampling to gradually reconstruct the data from pure noise

Training Objective: The network is trained using a simplified denoising objective, which is the mean squared error between predicted noise ε_θ and actual noise ε :

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^N \|\varepsilon_i - \varepsilon_\theta(x_{t_i}^{(i)}, t_i, y_i)\|^2$$

where:

- N is the number of samples in a training batch.
- ε_i is the actual noise added to the i^{th} sample
- $x_{t_i}^{(i)}$ is the actual noise added to the i^{th} sample
- y_i is the class label for the i^{th} sample

The difference for each prior assumption settings lying at the noise sampling process. In default stable diffusion, the noise is assume to come from Gaussian distribution. To evaluate the effect of prior distributions on scRNA-seq data generation, we modified the model by changing the prior assumption, using Log-normal and Student T distribution for comparison.

4.2.3.2 Benchmarking Model

For an overview comparison, we implemented two standard generative models: VAEs and GANs, compare with Stable Diffusion (using the prior with the best result). We then comparing with the evaluation metrics selected in section 4.2.5.

4.2.4 Synthetic Data Generation

The data generation process follows Algorithm 2 of Section 3.3.4.2 with the modification for each type of model. We choose cell types as our main condition for this thesis scope, but more cell ontology can be integrated into the model for a comprehensive generation. Data generation uses the learned reverse diffusion model to generate new synthetic data points, starting from random noise or a known prior distribution, and iteratively denoising:

- Input: random samples from the prior distribution
- Architecture: For each timestep $t = T, \dots, 1$, we predicted the noise component ϵ_θ and update the sample to remove noise.
- Output: generated data sample after all timesteps

4.2.5 Model evaluation

To evaluate the model, we adopt a part of the evaluation metrics from scDiffusion by Lou et al. [25]. With 2 types of metrics: statistical and non-statistical.

- Statistical Distance Metrics: we used two main metrics which are Energy Distance, Maximum Mean Discrepancy and Spearman SCC.
- Non-statistical Metrics: including UMAP representation and KNN AUC score.

Chapter 5

Experiment

5.1 Training setup

5.1.1 Dataset split

The dataset is stratified divided by labels into 80-20% proportion, with 80% using for training and 20% using for validation while training. The testing data is generated from the trained model and used to do model evaluation.

5.1.2 Model configuration

Below is the configuration table for the model

Hyperparameters	Value
Batch size	320
Learning rate	$1e^{-3}$
Max epochs	1000
Hidden dim	512
Embedding dim	128
Number of res block	6
Schedule type	linear
Number of timestep	1000
$\beta_{start}, \beta_{end}$	$[1e^{-5}, 0.02]$
Log-normal mu, sigma	(0,1)
Student T df	15

Table 5.1 Model configuration table

5.2 Result

5.2.1 Statistical Metrics

The experimental result for three statistical metrics: Energy Distance, Maximum Mean Discrepancy (MMD) and Spearman’s Rank Correlation Coefficient (SCC) are reported as below table.

	Energy Distance	MMD	SCC
Stable diffusion - Normal prior	0.256	0.004	0.996
Stable diffusion - Student’s t prior	0.255	0.005	0.995
Stable diffusion - Log-normal prior	6.45	0.511	0.837
VAEs	1.914	0.139	0.995
GANs	1.814	0.219	0.910

Table 5.2 Model performance comparison

In overall, the first class of model: Stable diffusion with Normal and Student’s t prior distribution setting outperformed the second class of model: Stable diffusion with Log-normal prior, VAEs and GANs in all three metrics.

- Energy Distance and MMD: these scores closer to 0 value in the first class, indicate the two distributions of generating data and real data are similar. This value is larger in the second class and achieved the highest value in Stable diffusion with Log-normal prior setting.
- SSC: this values close to 1 meaning the excellent preservation of rank relationships in the gene expression. For both first and second class of model, these value are generally close to 1, with Stable diffusion of Normal prior and Student ’s t prior performed slightly better, except for the Log-normal prior case where the value is considerably lower.

Stable diffusion with a Normal or Student’s t prior is the best-performing model, robust across all metrics. Meanwhile for Log-normal prior case, the results is significantly lower, likely due to its heavy-tailed and asymmetric nature of distribution. VAEs and GANs show reasonable performance but they are behind diffusion models in distributional similarity and structural fidelity.

5.2.2 Non statistical metrics

1. UMAP visualization

The visualization for generated and real is illustrated in the below figures. In overall, the visualization result is consistent with the statistical metrics reported in the previous section. The Stable diffusion model with Normal and Student's t prior have the ability to generate realistic samples with high fidelity, while the Log-normal settings fails to capture the realistic data structure.

Visualization for all cell types

Comparison between Generated and Real scRNA data: Normal prior

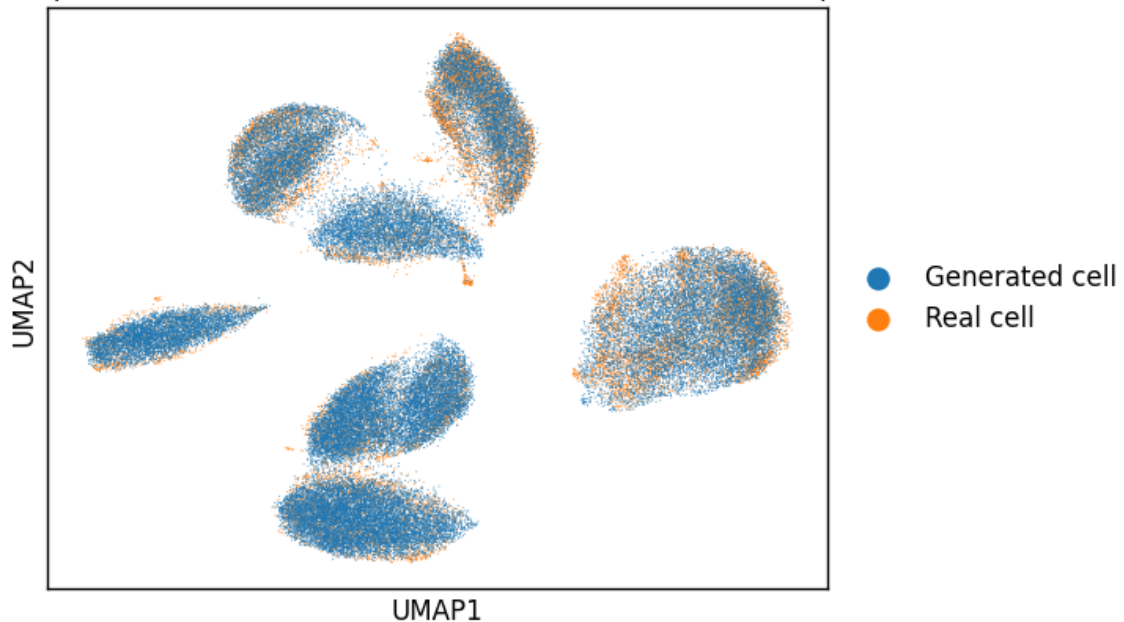


Fig. 5.1 UMAP of real data and generated data (Normal prior)

For the Stable Diffusion with Normal prior setting, the generated cells and real cells are nearly indistinguishable. This means the model has successfully captured both global and local gene expression patterns.

Comparison between Generated and Real scRNA data: Student T prior

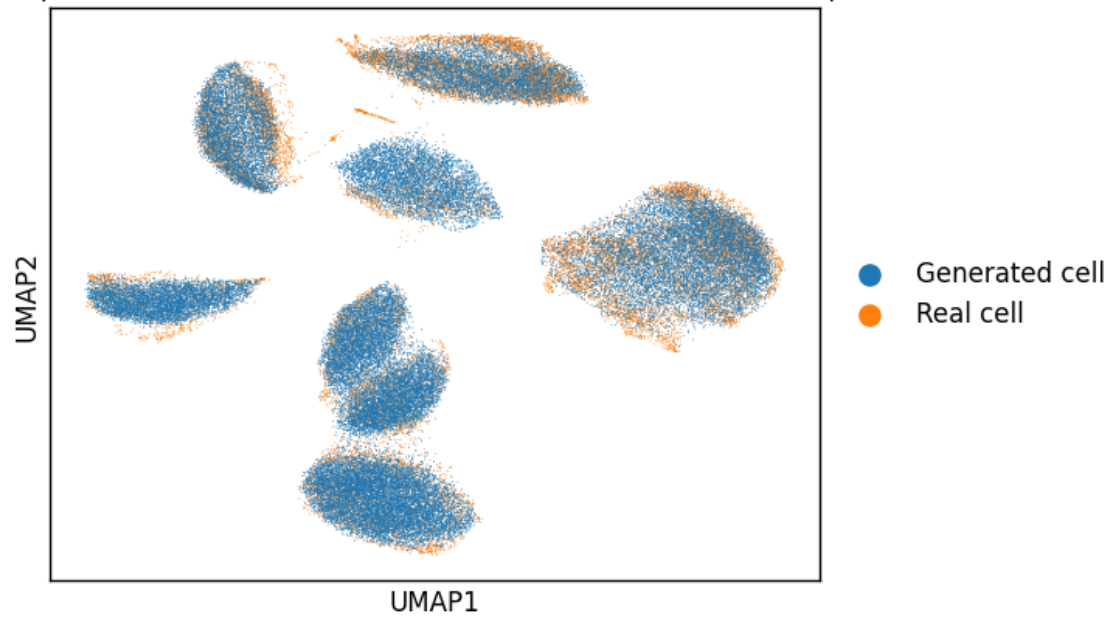


Fig. 5.2 UMAP of real data and generated data (Student T prior)

For the Stable Diffusion with Student's t prior setting, the result are similar with Gaussian distribution with high overlap between generated and realistic data. The clusters are more separate compares to the Gaussian.

Comparison between Generated and Real scRNA data: Log-normal prior

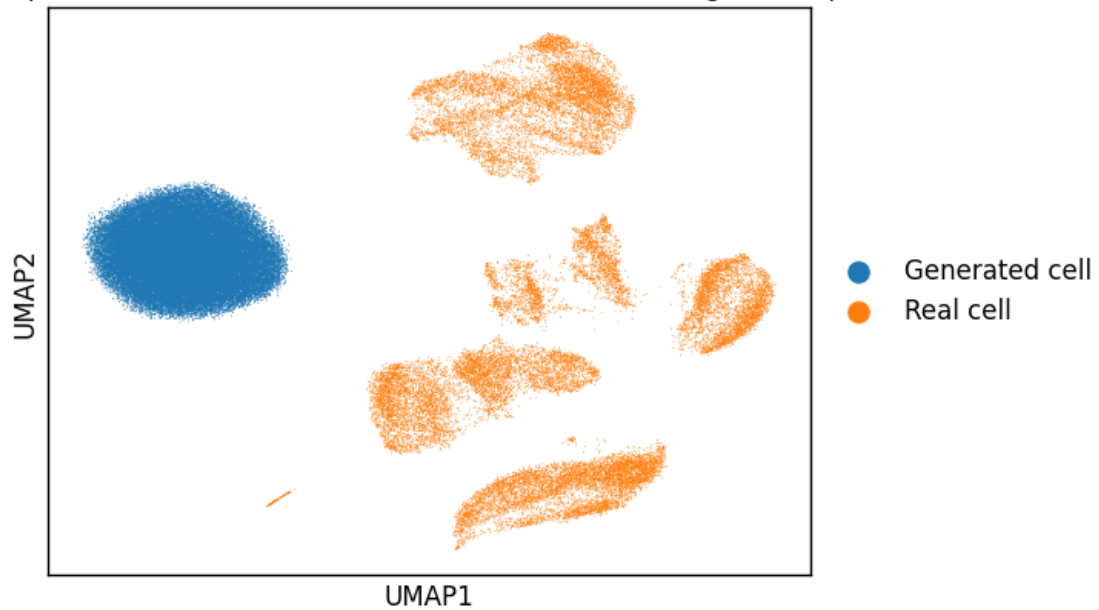


Fig. 5.3 UMAP of real data and generated data (Log-normal prior)

For the Stable Diffusion with Log-normal prior cases, the data generation failed with minimal overlap with real data. The distributions are disjoint, where the generated cells form a clearly separate dense cluster in one corner and the real data are spread throughout the data space.

Visualization for specific cell types

The UMAP visualization for some different cell types are also reported in the next figure, including B cell, T cell, Macrophage and Fibroblast, which are the core components of immune function and tissue homeostasis.

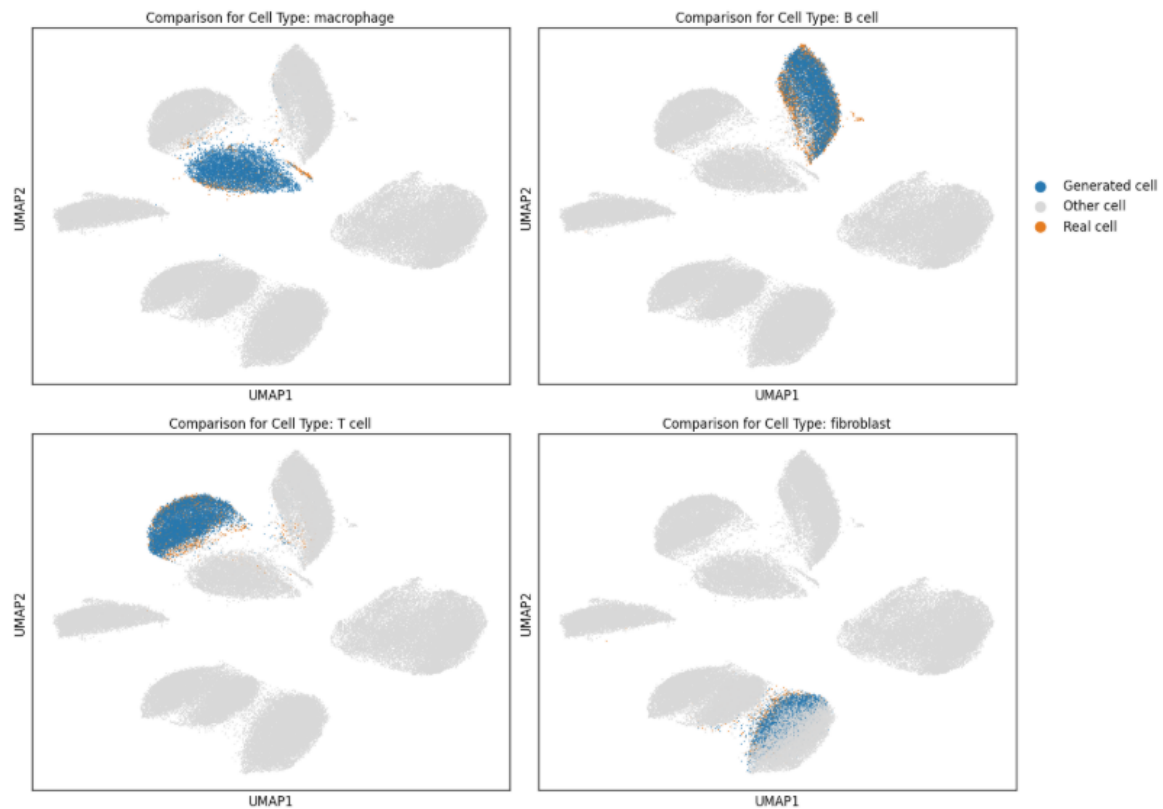


Fig. 5.4 UMAP of different cell types generated by Stable Diffusion (Normal prior)

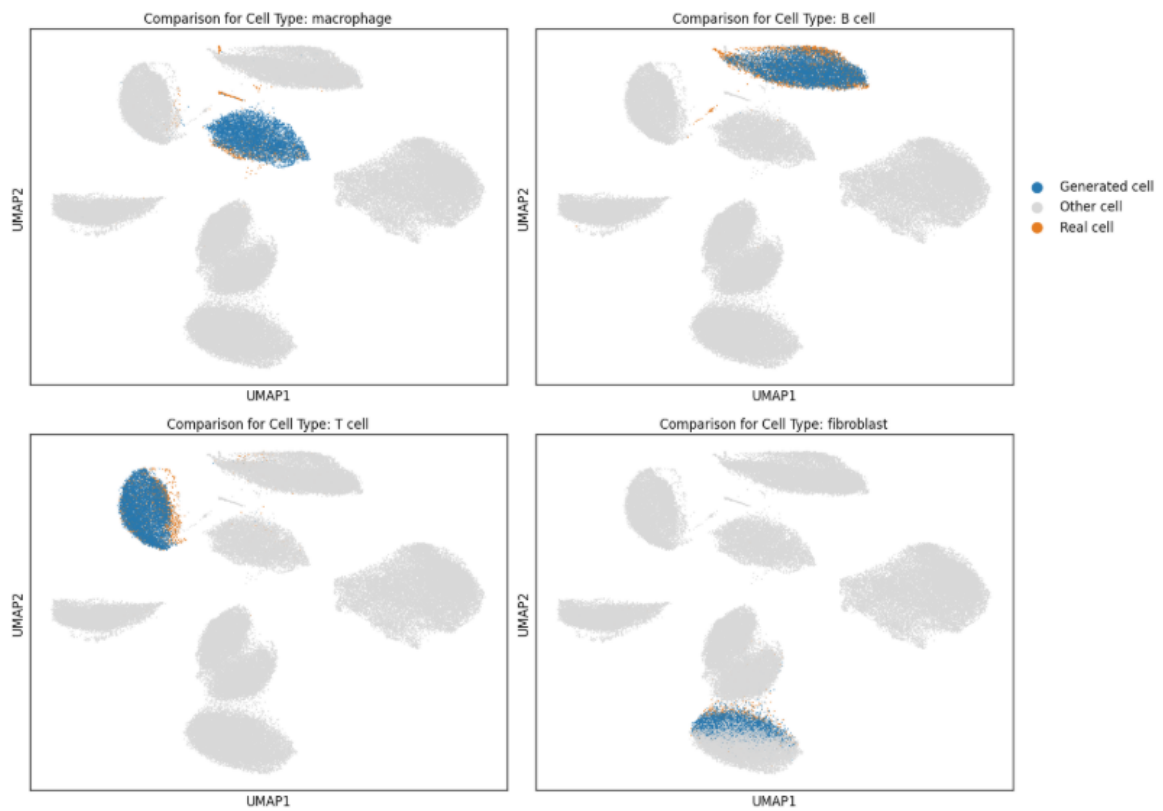


Fig. 5.5 UMAP of different cell types generated by Stable Diffusion (Student T prior)

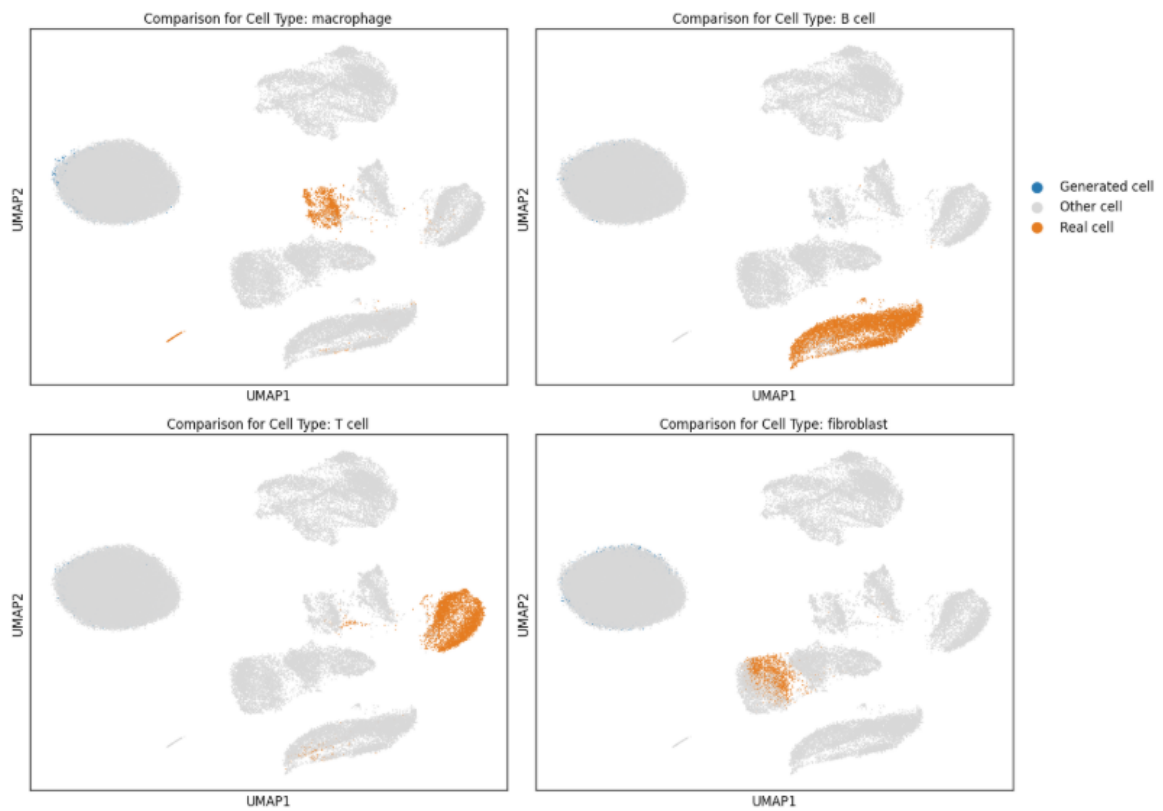


Fig. 5.6 UMAP of different cell types generated by Stable Diffusion (Log-normal prior)

2. KNN-AUC Score

The KNN-AUC scores for three prior assumptions and 10 cell types are illustrated using radar plots below. In general, Stable diffusion models with normal or Student's t priors produce significantly more realistic data than with a Log-normal prior.

For illustration purpose, the cell type name are encode as integer values. The full decoded table of cell type can be found in the below table.

Index	Cell type
0	B cell
1	T cell
2	basal cell of epidermis
3	chondrocyte
4	endothelial cell
5	fibroblast
6	keratinocyte
7	macrophage
8	mesenchymal progenitor cell
9	mesenchymal stem cell

Table 5.3 Cell type encoded table

KNN-AUC score for Stable diffusion with different prior settings

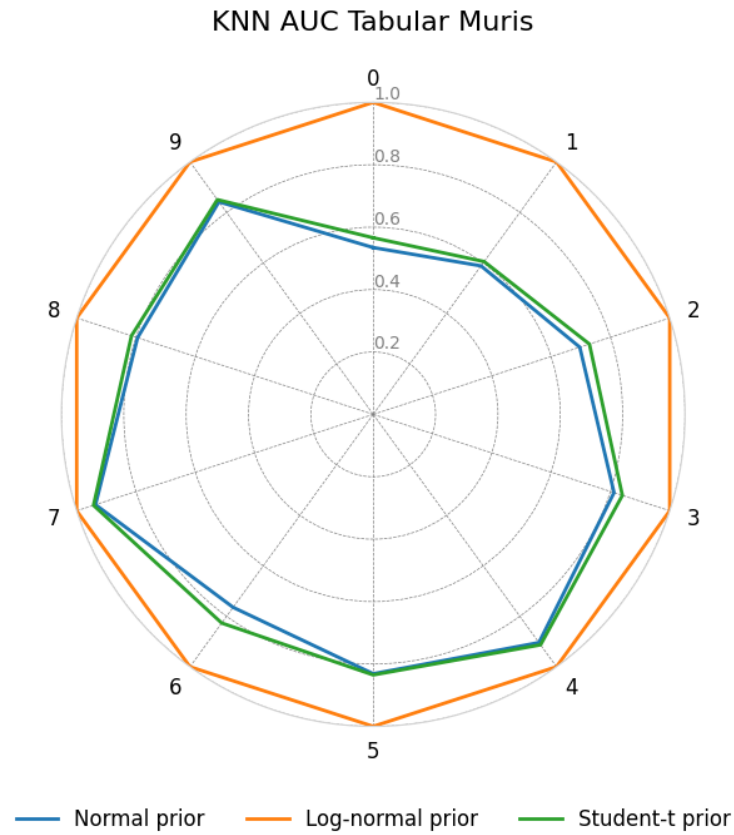


Fig. 5.7 KNN-AUC scores for different priors

- Normal and Student's t prior perform similarly and significantly better. Their AUC values are lower (closer to 0.5–0.7), meaning synthetic cells are less distinguishable from real ones, which means better realism and neighborhood preservation. They also maintain stable performance across all classes, indicating robustness in capturing diverse cellular structure.
- Log-normal prior consistently yields KNN-AUC values near 1.0 for all classes. This indicates that real and generated data are easily distinguishable, which is undesirable. It suggests the synthetic data lies far from the real distribution in local neighborhoods.

KNN-AUC score for different models

The KNN-AUC score for three models: Stable Diffusion (with Normal prior), VAEs and GANs are also reported in the below figure. In conclusion, Stable Diffusion model outperforms both VAEs and GANs in terms of KNN-AUC, confirming it better replicates the local distribution of real single-cell data.

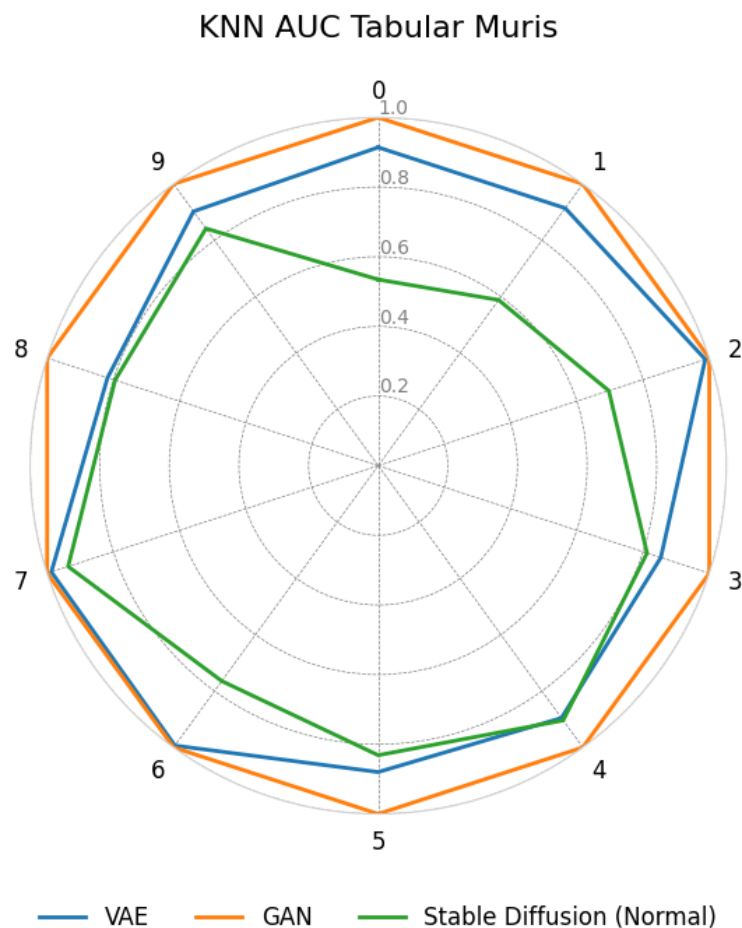


Fig. 5.8 KNN-AUC scores comparison of different models

- Stable Diffusion (with Normal prior) achieves the lowest KNN-AUC scores on most cell types (closer to 0.6), especially in mid-range classes (2–6), indicating superior ability to preserve local neighborhood structure.
- VAEs (blue) perform better than GANs, showing moderate KNN-AUC scores (0.7–0.9), though still not optimal.
- GANs perform the worst, with KNN-AUC values close to 1.0 across all classes, meaning they produce highly distinguishable (less realistic) data.

Chapter 6

Conclusion and Discussion

6.1 Conclusion

In conclusion, we have successfully integrated stable diffusion model for scRNA-seq data generation. In Chapter 1, we shown the motivation and objectives for this project and summarize all the relevant research together with the gaps and contribution of this thesis in Chapter 2. We then provided a solid background from biological, mathematics to ML/Deep Learning (DL) in Chapter 3 and a detailed description in material and pipeline of the thesis in Chapter 4. Finally we illustrated the effect of prior distributions assumption can affects tremendously to the quality of scRNA-seq data generation using Stable Diffusion model by comparing three types of distributions: Normal, Student's t and Log-normal across multiple metrics, using both statistical and non-statistical in Chapter 5.

Our research shows that Stable Diffusion with prior distributions as Normal and Student's t outperform other models such as VAEs and GANs across all metrics. With the assumption of Log-normal distribution for Stable Diffusion, the result is not promising and failed to generate realistic data. In overall, the choice of different prior can significantly impact the data set quality. We also successfully in implementing a classes of generative model from Stable diffusion to VAEs and GANs to applied in single cell data generation and bench-marking the result from each model.

6.2 Discussion

The thesis has provided a new discovery in diffusion model training where we can manipulated prior distribution to obtain a higher quality single-cell dataset. With this modification, we propose some further development strategy for this model:

- Conditional generation: instead of generating data only base on cell type, we could further expand this into generating data with many conditions such as time evolution to further explore the development of a cell (from a normal stage to cancer stage for instance). In order to do this effectively, we will need more implementation on classification guidance during the reverse process.
- Rare cell type generation: for this task we would need further training and model modification to balance between realistic and extreme cell type. Using a distribution such as Student's t could be a potential choice since the shape are similar to Gaussian distribution but with higher density value in the two tails. This can be a prospective direction to simulate rare cell type without generate too strong density value like Log-normal distribution, which cause the failure in training and data generalization.
- Evaluation metrics: to confidently evaluate the quality of the data generated, more effort in evaluation metric needed since the current evaluation metrics are dependence on the known data, which will not enough for evaluate the unseen data such as rare cell type or new cell type.

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